Machine learning-based adaptive optimization algorithm for Lipschitz continuity in biosignal telemetry

Información de publicación: Noorul Ameen, J; Irshad Ahamed, M; Senthil Mahesh, P. C; Banu Ahamed, Saahira.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In biosignal telemetry, signal integrity must be maintained against noise, artifacts, and transmission disturbances, especially in real-time health applications. In this relation, the present paper proposes the Machine Learning-Based Adaptive Optimization Algorithm (MLAOA) that ensures Lipschitz continuity across the various biosignal types, namely ECG, EEG, and EMG, by providing dynamic adjustment of signal processing parameters. The Algorithm incorporates supervised learning for signal behavior modeling with reinforcement learning for real-time adaptation. Experimental results on benchmark datasets show that MLAOA reduces the mean Lipschitz constant deviation to 2.7% (compared to 7.8% offered by more traditional methodologies), increases feature extraction accuracy by a margin of 14.3% and enhances signal-to-noise ratio significantly by an average of 9.6 dB. Concurrently, classification accuracy has been improved from 85.1 to 93.4%, whereas anomaly detection precision has been upgraded from 81.7 to 90.2%. With an average latency within 35 ms, MLAOA presents a lightweight and practical solution for real-time wearable and remote health systems, emphasizing the significance of incorporating machine learning with continuity principles for robust biosignal analysis.

ENLACES

Link to OCLC WorldCat

Generating effective ensembles for sentiment analysis

Información de publicación: Etelis, Itay; Rosenfeld, Avi; Weinberg, Abraham Itzhak; Sarne, David.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In recent years, transformer models have revolutionized Natural Language Processing (NLP), achieving exceptional results across various tasks, including sentiment analysis (SA). While current state-of-the-art approaches for SA predominantly rely on transformer achieving impressive accuracy levels on benchmark datasets, we hypothesize that strategically combining transformers with traditional NLP models can yield superior performance. In this paper, we introduce the hierarchical ensemble construction (HEC) algorithm, a novel greedy-based ensemble method that differs from traditional approaches (e.g., bagging, boosting, stacking) by iteratively building ensembles from scratch using simulated annealing to escape local optima. The key innovation of HEC lies in its empirically driven approach to ensemble construction. Through systematic experimentation, we discovered that selective inclusion of heterogeneous models outperforms traditional methods that assume all available models contribute positively to ensemble performance. Unlike conventional methods that use all available base-learners with different weights, HEC selectively identifies a minimal subset of complementary models that maximizes ensemble performance. Our empirical evaluation across eight widely-used SA datasets (including SST-2, IMDB, and YELP) demonstrates that HEC-based ensembles achieve a mean accuracy of 95.71%, yielding a statistically significant improvement ($p<0.05$) over both transformer-only ensembles and traditional ensemble methods. Specifically, HEC

reduces 26.61% of the performance gap between the best individual model and perfect classification, compared to only 11.02% for traditional methods. Additionally, we provide a comparative analysis with GPT-4 using zero-shot prompting, demonstrating that HEC outperforms GPT-4 in six out of eight datasets. Our results suggest that leveraging the complementary strengths of diverse model types through intelligent ensemble construction can advance the state-of-the-art in sentiment analysis.

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Trustworthiness in AutoML: a multi-query systematic review of frameworks, tools, ethical dimensions, and research gaps

Información de publicación: Albahri, A. S; Hamid, Rula A; Al-qaysi, Z. T; Chyad, M. A; Aljanabi, Mohammad; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Automated machine learning (AutoML) has emerged as a successful approach for addressing various fields by applying ML models to real-world problems through automation and increasing accessibility for organisations without specialised data scientists. The development of AutoML involves data preparation, feature engineering, model selection, hyperparameter tuning, and model deployment. However, challenges remain in ensuring these models are trustworthy, explainable, fair, privacy preserving, and robust. Therefore, how literature addresses the automated process and the trustworthiness considerations remains unclear. This paper provides a comprehensive overview of the contributions of traditional AutoML methods and those specific to trustworthy-based AutoML, exploring critical aspects of design and development within these frameworks. This aim has been achieved using two keyword queries, capturing research across traditional AutoML methods and newer trust-focused innovations. We conducted a structured search and filtering process across five major databases—ScienceDirect, IEEE Xplore, Scopus, the Association for Computing Machinery, and Web of Science—between 2019 and 2025. We applied a multi-query systematic review process and analysed 86 peer-reviewed studies (71 traditional and 15 trustworthy focused) through qualitative synthesis and taxonomy development. We identified nine categories from 71 retrieved studies to formulate a specific taxonomy of traditional AutoML. The identification of trustworthy AutoML innovations was attributed to only 15 studies in three categories. This study is coupled with an in-depth discussion of new taxonomies, motivations, challenges, and recommendations. To evaluate these studies, we incorporated performance indicators such as method type, use of explainability tools, fairness measures, robustness handling, and application domains. The findings highlight the relative scarcity of research specifically targeting trustworthiness in AutoML, indicating key gaps in the literature. Our work highlights six research gaps related to AutoML from different perspectives. Furthermore, it critically assesses the state-of-the-art advancements of trustworthy artificial intelligence requirements in AutoML literature. This study analyses existing AutoML methods using dataset examples, then proposes protection strategies against adversarial attacks, and how to implement a multicriteria decision making (MCDM) approach enhanced by expert input. The results are discussed using both qualitative synthesis and structured comparison tables, highlighting specific capabilities, limitations, and opportunities across included studies. Researchers will benefit from this examination because it demonstrates techniques to build trustworthiness in AutoML technology, which will help advance future trustworthy AutoML investigations.

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Editorial: data science and AI for marine science and the blue economy

Información de publicación: Candela, Leonardo; Pagano, Pasquale; Bi, Hongsheng; Schaap, Dick.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This editorial introduces the Special Collection “Data Science and AI for Marine Science and the Blue Economy” published in the International Journal of Data Science and Analytics. The collection explores how data-driven and AI-enabled approaches are advancing marine research, supporting operational monitoring, and enabling evidence-based decision-making across blue economy domains. The guest editors summarize the motivations for this initiative, briefly present the contributions included in the issue, and outline emerging themes and future perspectives in this evolving interdisciplinary field.

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A comprehensive review and future directions on blockchain-integrated deep federated learning for privacy-preserving healthcare applications

Información de publicación: Karmode, Sayali; Shahade, Aniket K.

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RESUMEN (ENGLISH)

This review examines how Federated Learning (FL) and Blockchain (BC) can work together to support privacy-preserving, auditable and scalable AI in healthcare, with a focus on Ayurveda. Using a structured database search and clear inclusion and exclusion criteria, We reviewed 87 papers mainly from 2020 to 2025, plus few of foundational works published earlier on Federated Learning in healthcare, Blockchain—Federated Learning frameworks, Deep Learning (DL) integrated with Federated Learning and Blockchain, and AI-based work in Ayurveda, Explainability in Healthcare AI guided by five research questions. The findings show that FL enables multi-centre model training without sharing raw data, while BC adds tamper-evident logging, fine-grained access control and incentive mechanisms. Integrated strong privacy methods and consensus choices create trade-offs in latency, energy use, scalability and system complexity, which must be balanced against the needs of each healthcare setting. The Ayurveda-focused analysis shows that existing AI systems are mostly small-scale, centralised and weak in provenance and privacy, so issues of data scarcity, robustness and trustworthy deployment remain unresolved. As a future direction, the review proposed a Proof of Authority (PoA) style, reputation and gradient aware with

Deep Federated Learning (DFL) integrated Blockchain framework for Prakriti and Dosha-informed personalised Ayurvedic care, offering a focused roadmap for privacy preserving compliance adhering yet accurate deployment.

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A new take on optimal vaccine prioritization: when age-based strategy is efficient?

Información de publicación: Gonçalves, Miguel; Fierens, Pablo Ignacio; Rêgo, Leandro Chaves.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The rapid spread, high mortality rate, and hospital overcrowding made the SARS-CoV-2 virus a significant challenge for humanity. The emergent situation urgently called for investments in vaccine development as a critical measure to mitigate this issue. However, the financial costs and time required for vaccine development posed considerable obstacles to achieving comprehensive vaccination coverage. Therefore, understanding the most effective immunization strategy was paramount to contain the virus's spread and optimize resource allocation in public policies. Age-based prioritization was applied in many regions, and one of the objectives of this work is to study under which circumstances such a strategy is efficient. Our results show that age-based vaccination is a poor strategy for disease eradication, although it has a fair performance in diminishing the number of deaths. Furthermore, other schemes, such as PageRank-based immunization, are much better options in general. We believe these results are of relevance, not only for the past SARS-CoV-2 pandemic, but also for future epidemic outbreaks.

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Constraint-biased transformers: attention bias injection for rule-compliant course recommendation

Información de publicación: Ibrahim, Ahmed.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Transformer-based sequence-generation models have achieved remarkable success across a range of NLP tasks. However, ensuring they adhere to complex, domain-specific constraints remains a major challenge—especially in settings like academic planning and workflow automation. In this paper, we introduce a constraint-based Transformer that promotes rule-adherence via bias injection. During training, each input sequence is augmented with canonical, rule-compliant examples whose positional embeddings are integrated into the model's attention bias matrices, steering the Transformer toward valid orderings without sacrificing generalization. We demonstrate this approach on an academic-planning task,

evaluating performance on a held-out synthetic dataset using a constraint-adherence metric (the percentage of outputs satisfying all domain rules) alongside standard language model metrics. Our model achieves 87.40% adherence—surpassing a standard Transformer (87.10%) and an LSTM model (85.50%)—while matching their perplexities (1.28) and Top-5 accuracies (98.0%). These results show that bias injection can effectively bridge rule-based and data-driven methods, delivering more reliable recommendations. Future work will explore adaptive bias schedules and extensions to heterogeneous rule sets.

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A two-layer multivariate decomposition framework for spatiotemporal tourism demand forecasting based on Internet search and city synergy factors

Información de publicación: Zhang, Haina; Liu, Wenzheng; Shi, Jie; Li, Hongtao; Sun, Shaolong.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Accurate forecasting of tourism demand plays a critical role in optimizing the allocation and management of tourism resources. However, identifying tourist intentions and trends remains challenging due to the complexity and multidimensional nature of influencing factors. To address these challenges, this study integrates Internet search indices with tourist arrival data to trace the spatiotemporal dynamics of tourism demand in a target city and its surrounding regions. A novel two-layer multivariate decomposition ensemble framework is proposed to reduce data complexity and achieve comprehensive analysis and feature extraction. The framework employs a two-step process: multivariate empirical mode decomposition is used to extract intrinsic mode functions (IMFs) that represent distinct features, followed by multivariate variational mode decomposition applied to high-complexity IMFs to uncover deeper interrelationships. These IMFs are then predicted using a multivariate gated recurrent unit, and the final forecasting result is obtained by aggregating the predictions. Empirical evaluations on two case studies, Nanjing and Haikou, demonstrate that the proposed framework achieves superior predictive accuracy compared to baseline models. The results highlight the effectiveness of integrating spatiotemporal data with advanced decomposition methods to uncover latent patterns in tourism demand. Furthermore, we provide actionable recommendations for tourism policymakers and stakeholders, offering insights into resource allocation, congestion management, and sustainable tourism development.

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Toward a unified agentic framework for regime-aware portfolio optimization with LLM signals

Información de publicación: Mantshimuli, Lamukanyani Alson; Muteba Mwamba, John Weirstrass.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Most portfolio optimization frameworks assume static objectives and constraints, making them fragile under regime shifts, transaction frictions, and evolving information. Existing LLM-based methods focus on signal generation without governing execution or constraint elasticity, while reinforcement learning approaches often lack transparency and cost discipline. This lack of a unified, interpretable architecture hinders adaptability and accountability during live rebalancing. We address this gap by introducing an agentic portfolio optimization framework that integrates regime-aware convex optimization, LLM-derived sentiment and uncertainty features, and a constrained reinforcement learning controller in a closed loop. The agent senses market and news data, infers regimes, and dynamically adjusts objectives, risk budgets, and position limits, while enforcing friction-aware execution through Sharpe-gated trade activation, partial rebalancing, and turnover budgeting. In a 50-stock S & P 500 portfolio tested under walk-forward evaluation (2021–2025 Q1), agentic portfolios consistently outperform non-agentic benchmarks, with Sharpe ratio gains of up to +0.373 (NSGA-3), persisting net of transaction costs and alongside lower turnover. These results highlight the value of combining forward-looking signals, regime-conditioned intent, and disciplined execution within a unified transparent, adaptive agentic architecture. Future research should explore multi-asset extensions, richer textual and alternative data for regime inference, explainability metrics for LLM-driven signals, and hybrid architectures blending deterministic control with selective policy learning for extreme market conditions.

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Charting the evolution of neuro-symbolic AI in cybersecurity: a scientometric perspective

Información de publicación: Shama; Jain, Sarika.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Neuro-symbolic artificial intelligence (NSAI) integrates neural learning with symbolic reasoning to address complex cybersecurity challenges through interpretable and adaptive solutions. This study presents the scientometric mapping of NSAI research in cybersecurity, analyzing publications from Scopus database spanning 2016–2025. Through keyword co-occurrence, bibliographic coupling, and systematic literature review, we identify four thematic clusters and reveal dominant integration paradigms. Our findings show rapid field growth, with Learning-for-Reasoning architectures as the predominant approach. Network intrusion detection and malware analysis emerge as mature domains, while autonomous cyber defense and IoT security remain underexplored. Significant research gaps remain in developing computational efficiency benchmarks, standardized evaluation frameworks, and explainability mechanisms. This integrated approach provides strategic insights for advancing NSAI-driven cybersecurity research and practice.

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Covertness-aware graph neural networks for misinformation detection in e-commerce

Información de publicación: Narayan, Abhay; Tharun, Dameera; SD, Madhu Kumar; Chacko, Anu Mary.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Product reviews on e-commerce platforms strongly influence consumer purchasing decisions but are increasingly targeted by fraudulent actors seeking to manipulate public perception. This paper presents COVERT-GNN, a novel graph-based framework that integrates metadata, multi-relational structures, and Graph Neural Networks (GNNs) to detect misinformation in user reviews. Unlike existing approaches that rely on static or handcrafted features, COVERT-GNN introduces a vertex-aware covertness mechanism that quantifies both feature-level mimicry and relation-level concealment to reveal fraudsters who intentionally imitate benign users. The framework further incorporates relation-aware neighborhood aggregation and a reinforcement-driven neighbor selection strategy, enabling the model to adaptively capture latent behavioral dependencies across users, items, and reviews. Extensive experiments on the Yelp and Amazon datasets demonstrate that COVERT-GNN achieves strong and consistent performance, obtaining an AUC of 75.52%, Recall of 69.28%, and F1-Macro of 63.70% on Yelp, and an AUC of 93.12%, Recall of 88.03%, and F1-Macro of 92.46% on Amazon. These results confirm the effectiveness of COVERT-GNN in identifying covert misinformation patterns that mimic genuine reviews, offering a robust and scalable solution for reliable fraud detection in e-commerce environments.

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Emotion-aware multimodal framework with similarity-guided gating for disaster misinformation detection

Información de publicación: Wang, Feifan; Zhang, Feng; Wang, Jingge; Yan, Haonan; Li, Chaohao.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The rapid dissemination of disaster-related misinformation on social media poses serious risks to public safety and data-driven crisis response. Detecting such content remains challenging because misleading posts often combine emotionally charged text with reused, irrelevant, or weakly related images. This multimodal and affective complexity limits the effectiveness of text-only models and reduces the reliability of existing multimodal approaches. This paper proposes an emotion-aware multimodal framework for disaster misinformation detection that jointly models text, images, emotional signals, and image–text similarity. The approach utilizes cross-modal attention and similarity-guided gating to regulate visual contributions based on semantic alignment, thereby enhancing robustness to misleading images. Experiments on the large-scale Fakeddit dataset show that the proposed method achieves 0.943 accuracy and 0.942 F1-score, outperforming multimodal baselines. When transferred to the Crisis dataset, the model attains 0.895 accuracy and 0.873 F1-score, indicating robust cross-domain generalization under low-resource disaster settings. Additional robustness

experiments confirm stability under modality degradation, while analysis of input features highlights the complementary roles of domain-specific emotional cues and domain-invariant image–text inconsistency. These results suggest the value of combining affective signals with alignment-aware multimodal analysis for reliable detection of disaster misinformation.

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Bibliometric analysis and visualization of global trends in cyber-attacks and defenses

Información de publicación: Chuang, Yung-Ting; Chuang, Wen-Yuan.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

As cyber threats grow in complexity, understanding the global research landscape of cyber-attacks and defenses has become critical for ensuring digital resilience. Despite rapid technical advancements, a significant research gap exists in the integrated analysis of technical innovations and their socio-economic implications. This study aims to bridge this gap by conducting a comprehensive bibliometric analysis of 11,561 articles indexed in SCIE and SSCI from 1998 to 2023 using CiteSpace. By comparing technical developments in SCIE with socio-economic trends in SSCI, this research provides a holistic map of the field's evolution. Publication counts, citations, research fields, and journal impact factors were assessed, and co-authorship networks and co-citation patterns were analyzed. Key findings reveal a significant geographic shift, with a notable geographic shift toward emerging research hubs in the Middle East and Asia, alongside increased institutional activity from entities such as the Egyptian Knowledge Bank. Our analysis identifies 2018 as a pivotal year where AI literature emerged due to advancements in deep learning frameworks and datasets. We found that research has matured from reactive "intrusion detection" toward proactive, AI-driven defense mechanisms in SCIE, while SSCI research emphasizes human factors and governance. The results highlight that cybersecurity has evolved into an interdisciplinary challenge where technical breakthroughs are increasingly integrated with legal and organizational frameworks. This study provides a strategic roadmap for future research, emphasizing that staying ahead of threats requires balancing technical precision with ethical and behavioral insights.

ENLACES

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Unveiling the dynamics of student physique: a time-varying graph kernel approach for collegiate fitness assessment from wearable data

Información de publicación: Diliya, Dilixiati; Wen, Nan.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Forecasting complex human health outcomes like college student fitness demands models that handle nonlinear relationships while grasping the evolving links among behavior indicators over time. Existing machine learning methods often stumble under assumptions of linearity, static setups, or feature independence. These fail to capture real-world dynamics mirrored in wearable sensor data. To break free, we introduce a fresh modeling approach: Dynamic Graph-Kernelized Regression (DGKR). This end-to-end unified framework delivers a full "dimensional upgrade" to tackle three core hurdles. It evolves the Health Behavior Network (HBN) from a static snapshot into a time-varying graph sequence. It lifts regression via the kernel trick into an infinite-dimensional Reproducing Kernel Hilbert Space (RKHS) for arbitrary complexity. And it weaves in robustness through personalized self-attention and quantile regression for adaptive handling and uncertainty measures. We applied DGKR to a real college health dataset (FAW) and several general time-series regression tasks from the UEA & UCR archive. Broad experiments show DGKR tops prediction accuracy across the board, statistically outpacing 11 strong baselines spanning diverse categories, including XGBoost, LSTM, and state-of-the-art dynamic graph networks.

ENLACES

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A self-matching training method with annotation embedding models for ontology subsumption prediction

Información de publicación: Shiraishi, Yukihiro; Kaneiwa, Ken.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Recently, ontology embeddings representing entities in a low-dimensional space have been proposed for ontology completion. However, the ontology embeddings for concept subsumption prediction do not address the difficulties of similar and isolated entities and fail to extract the global information of annotation axioms from an ontology. In this paper, we propose a self-matching training method and two ontology embedding models: **In**verted-index **M**atrix **E**mbedding (InME) and **Co**-occurrence **M**atrix **E**mbedding (CoME). The two embeddings capture the global and local information in annotation axioms by means of the occurring locations of each word in a set of axioms and the co-occurrences of words in each axiom. The self-matching training method increases the robustness of concept subsumption prediction for ontologies with sufficiently rich annotation axioms, especially when predicted superclasses are similar to subclasses or correspond to isolated entities. Our evaluation experiments show that the self-matching training method with InME outperforms the existing ontology embeddings for the GO and FoodOn ontologies and that the method with the concatenation of CoME and OWL2Vec* outperforms them for the HeLiS ontology.

ENLACES

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Beyond sentiment: knowledge graph-driven financial market forecasting with large language models-extracted enterprise relations and adaptive residual networks

Información de publicación: Chen, Weisi; Zheng, Jiaxin; Li, Yuan; Zhang, Xu; Wang, Han; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Financial markets exhibit high volatility driven by multi-factor coupling; yet traditional forecasting methods have the following limitations: insufficient integration of heterogeneous data, a lack of explicit modeling for dynamic relationships between enterprises, and hence generating ambiguous or even incorrect sentiment polarity, leading to inferior forecasting results. To address these issues, this paper proposes a knowledge graph (KG)-driven financial market forecasting framework that integrates large language models (LLMs) and an adaptive deep learning model. First, it combines technical indicators, specifically including the Fibonacci indicator, and news sentiment to enhance the market perception. Second, an LLM-based extraction module parses unstructured corporate news to construct a lightweight KG, defined by a simplified schema that focuses exclusively on dynamic competitive/cooperative binary relationships without redundant attributes. This structure enables efficient automated construction through prompt engineering with no reliance on complex ontologies or extensive manual annotation. Third, the T5-Relation-Aware Attention (T5-RAA) sentiment model incorporates KG-derived relationship bias into the attention mechanism, enabling distinction between “cooperative positive events” and “competitive negative events”. Finally, the Bidirectional Long Short-Term Memory (BiLSTM) with Attention and Layer-wise Residual Connections (BA-LRC) prediction model takes the sentiment result from the T5-RAA model and employs residual blocks to preserve temporal features and stabilize training. Experiments on nine stocks in various markets demonstrate that the integration of the KG improves the F1-score of sentiment classification by 16%; the BA-LRC model achieves state-of-the-art prediction performance, with an average coefficient of determination (R^2) of 0.9695 and a root mean square error (RMSE) of 2.0861, outperforming baseline models in continuous price forecasting tasks. While maintaining competitive trend direction accuracy comparable to top baselines, our model significantly reduces regression error, demonstrating superior capability in capturing precise market valuations. The trading simulation based on the model’s predictions demonstrates robust practical application value. This study has established a knowledge-driven forecasting paradigm, deeply integrating the relational semantics of the KG into sentiment analysis and time-series prediction, solving the problem of sentiment polarity bias caused by the lack of relational context; meanwhile, it provides a low-cost and adaptive solution for the construction of dynamic KGs in the financial field and enhances the model’s generalization ability across different markets and high-frequency scenarios.

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A neural network-based system for automated refrigeration alarm management in food retail

Información de publicación: Ferrer-Cid, Pau; Albets-Chico, Xavier; Barcelo-Ordinas, Jose M; Garcia-Vidal, Jorge.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The increasing use of refrigeration monitoring platforms in the food retail sector is crucial for enhancing operational efficiency and mitigating food waste. These systems are essential for ensuring food safety and quality, as they alert personnel to abnormal temperature readings, which are often caused by equipment malfunctions or operational issues. However, the resulting high volume of alarms poses a significant challenge, requiring tailored management strategies and continuous supervision by expert staff. This article presents an intelligent alarm labeling system based on neural networks that evaluates various alarm dimensions. The proposed system effectively detects recurrent patterns and determines a device's status based on its temperature measurements. The system was validated using a massive monitoring dataset from different supermarkets in Catalonia, Spain. The results show that the system can detect recurrent alarms with 98% accuracy and correctly identify device states with 93% accuracy, enabling more effective temperature alarm management.

ENLACES

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A review of security threats and privacy issues in federated learning

Información de publicación: Agrahari, Abhishek Kumar; Dinker, Aarti Gautam; Singh, Rajendra Bahadur.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Federated learning (FL) has emerged as a promising paradigm for collaborative model training while preserving data privacy by keeping sensitive data localized at client devices. Instead of sharing raw data, participating clients exchange model updates, which can reduce communication overhead compared to centralized learning, depending on the training configuration and data distribution. However, this decentralized setting introduces significant security and privacy vulnerabilities. This review presents a structured and threat-based analysis of security risks in federated learning, focusing on backdoor attacks, Byzantine attacks (including Sybil-based mechanisms), and adversarial attacks. Following a systematic literature review of peer-reviewed and high-impact studies published between 2018 and 2025, we examine attack strategies, defense mechanisms, and their underlying assumptions. A comparative synthesis highlights critical trade-offs among robustness, privacy, computational cost, and convergence behavior, particularly under non-IID data distributions and resource-constrained environments. The paper proposes a taxonomy that clarifies the relationships between attack phases, threat models, and defensive strategies, and identifies open research challenges such as adaptive adversaries, privacy–robustness conflicts, and scalable defenses for real-world deployments in domains including healthcare, IoT, and edge intelligence. This work aims to provide both a consolidated reference and a roadmap for designing secure and trustworthy federated learning systems.

ENLACES

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Flexible priors for cross-collection topic modeling: an approach with generalized Dirichlet and Beta-Liouville distributions

Información de publicación: Luo, Zhiwen; Fan, Wentao; Amayri, Manar; Bouguila, Nizar.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Cross-collection topic models (ccTMs) aim to learn both shared and collection-specific topics across multiple document collections. Many ccTMs are built on cross-collection Latent Dirichlet Allocation and use Dirichlet priors, which impose a restrictive covariance structure on topic proportions and topic-word distributions. In addition, several ccTMs rely on exact topic-index alignment across collections, which can be too rigid when collections differ in theme coverage. We propose GD-BL ccTM, a cross-collection topic model that uses a Generalized Dirichlet (GD) prior for document-topic proportions and a Beta-Liouville (BL) prior for topic-word distributions. These priors allow a less restrictive covariance structure, improving modeling of topic co-occurrence and word dependencies. GD-BL ccTM also relaxes strict alignment by separating shared global topics from collection-specific topics and using a token-level selector to route each word to the shared or collection-specific component. We evaluate GD–BL ccTM on real-world datasets for comparative text mining and cross-domain classification. Results show consistent improvements over competitive baselines in perplexity, topic coherence, and classification accuracy, supporting the benefits of flexible priors and relaxed alignment in cross-collection settings.

ENLACES

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Associative modeling of Chinese character stroke sequences combining transformer and geometric constraints

Información de publicación: Yang, Yan.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This research introduces a combined deep learning framework for classifying Chinese stroke sequences using multi-scale spatial encoding, hierarchical attention modeling, and geometric constraint learning. Unlike previous Swin Transformer-based stroke classification models, this study presents a novel hybrid framework that combines U-Net's multi-scale spatial encoding with Swin Transformer's hierarchical attention, explicitly capturing both local and global dependencies in handwritten stroke sequences. Each stroke sample is encoded into a 2D trajectory-density feature map, rather than raw coordinate sequences, to retain spatial continuity, temporal progression, and fine-grained geometric structure. This approach ensures the joint learning of spatial–temporal patterns while preserving stroke continuity and geometric fidelity, providing a clearer representation for the model to classify stroke sequences accurately. The U-Net framework enables the

extraction of multi-resolution spatial features, while the Swin Transformer captures long-range contextual dependencies with shifted-window self-attention. A geometric constraint loss term is included to capture curvature smoothness, directionality consistency, and structural fidelity, addressing the challenges of visual similarity and significant variability of handwritten strokes. Experiments conducted with four main stroke classes—heng, shu, pie, and na—demonstrated strong classification performance with 98.57% accuracy, an average precision of over 0.994, and an AUC of over 0.995. Experiments conducted with the Handwritten Chinese Stroke 2021 dataset demonstrated strong classification performance with 98.57% accuracy, surpassing state-of-the-art methods such as Swin Transformer-based stroke classification models and CNN-based models for Chinese character recognition. Evaluations with the confusion matrix, ROC curves, precision–recall curves, and error rate (FPR/FNR) metrics each establish the robustness and generalizability of the model across different handwriting styles. These findings indicate that the framework effectively captures stroke-level spatial–temporal patterns and provides a robust basis for downstream applications, including character reconstruction, handwriting analysis, digital calligraphy, and intelligent writing-education systems.

ENLACES

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Knowledge graph and semantic representation enhancement for collaborative knowledge building

Información de publicación: Chen, Jie; Li, Jiazheng.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In the field of knowledge building, “Discourse” plays a central role. If real-time and precise classification, as well as effective monitoring and intervention, can be achieved for the various discourses generated during the process of online collaborative knowledge building, it will greatly promote “idea improvement”. However, knowledge building systems rely heavily on the semantic recognition and judgment of natural language. Thus, existing knowledge building systems augment the context through external resources such as knowledge graphs and model entities based on their interrelationships. Nevertheless, these methods overlook the rich intrinsic information within entities. To address this issue, we introduce a knowledge graph-based enhanced entity representation learning framework. This framework leverages knowledge graphs and pre-trained language models to improve the semantic understanding of entities. It effectively captures the temporal information of entities in conversations using positional encoding. Through this framework, the accuracy of discourse classification in the knowledge building process is significantly enhanced. Comparative and ablation experiments demonstrate that the SVM algorithm outperforms other methods, achieving an accuracy of 92% and an F1 score of 87% in discourse classification. This method holds significant value in the context of online collaborative knowledge building, pedagogy and technology. It effectively supports the development of process evaluation and precisely regulates learners’ knowledge building processes. Furthermore, it promotes the continuous enhancement of discourse, providing robust support for the high-quality development of online collaborative knowledge building instruction.

ENLACES

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Vision-based wafer inspection in semiconductor manufacturing: a case study on scratch defect detection using synthetic data and YOLO models

Información de publicación: Peñaloza, Ramón Jesús Sánchez; Moghadam, Armin; Davoudi Kakhki, Fatemeh.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Scratch-type surface defects are among the most critical in wafer fabrication. Even minor scratches can result in rejected dies, substantial yield loss, and downstream reliability failures, making their early and precise detection essential for cost-effective production and long-term device performance. Given the scale and speed of modern semiconductor manufacturing, manual or traditional inspection methods are insufficient to reliably detect such subtle surface defects, underscoring the urgent need for intelligent, automated inspection systems capable of real-time, high-precision detection. However, developing such systems remains challenged by limited access to diverse, labeled defect datasets and the high cost of manual annotation. To overcome this, we introduce a novel, fully synthetic-data-driven inspection framework that bridges virtual simulation with real-world deployment, without requiring any annotated real-world training data. A high-fidelity 3D simulation pipeline was developed using Blender to procedurally generate thousands of labeled wafer images featuring scratch-type surface defects under varying lighting, texture, and geometric configurations. Concurrently, a physical optical inspection system was constructed to capture wafer images under controlled conditions, serving as the experimental validation platform. YOLOv8m, YOLOv10m, and YOLOv11m object detection models were trained exclusively on the synthetic dataset and directly evaluated on real wafer images obtained from the custom inspection setup. Results demonstrated that YOLOv8m and YOLOv11m achieved detection accuracy of F1 = 1.00 in wafer presence classification and F1-scores exceeding 0.96 in scratch defect localization, indicating that models trained on synthetic images can achieve strong performance on real wafer images within the tested physical environment. This two-stage evaluation pipeline eliminates the need for costly manual annotation while preserving high detection fidelity. The proposed framework combines synthetic data generation and YOLO-based deep learning to address critical data bottlenecks in industrial defect detection.

ENLACES

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A new optimization model for regularization in CNNs using spatial correlation and guided by particle Swarm optimization

Información de publicación: Ettaouil, Asmae; EL Khaoulani, Rachid; Joudar, Nour-Eddine.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Convolutional neural networks learn discriminative feature structures that are crucial for complex tasks, though their high complexity makes them susceptible to overfitting, particularly when dropout is applied to convolutional layers, where spatially correlated activations limit their effectiveness. Although structured regularization methods such as DropBlock address this issue, they do not take into account semantic information and may drop discriminative regions essential for classification. Thus, a structured form of dropout is needed that considers these important spatial correlations. In this case, we propose a novel regularization method, called PSO-Drop, which introduces a new optimization model for regularization. To solve this optimization problem, we define a new particle structure, where each particle represents a candidate solution in the particle swarm optimization (PSO) framework, explicitly preserving discriminative regions with strong spatial correlations. Extensive experiments demonstrate that PSO-Drop improves classification accuracy and outperforms state-of-the-art regularization methods.

ENLACES

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An uncertainty and conviction-aware attention model for automatically estimating reviewer confidence from peer review texts

Información de publicación: Bharti, Prabhat Kumar.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Peer review is the primary mechanism for ensuring the quality and integrity of scholarly publications. Reviewers are often asked to explicitly report their level of confidence when evaluating a manuscript, particularly in conference reviewing. In many cases, especially for lengthy or complex submissions, editors and area chairs rely on these self-rated confidence scores to resolve conflicting reviews and make decisions in borderline cases. However, reviewers' self-annotations of confidence do not always align with the impressions conveyed by the review text itself, leading to potential inconsistencies. As a result, such self-reported confidence is frequently miscalibrated and may fail to accurately reflect a reviewer's true conviction regarding the merits of a paper. Review texts often contain *uncertainty-related linguistic cues*, such as hedging expressions (e.g., *maybe, seems, might*), which can indicate ambiguity, limited supporting evidence, or cautious reasoning. Editors and area chairs must implicitly account for these cues when interpreting reviews and making final decisions. In this work, we propose a generic attention-based framework to computationally estimate reviewer confidence directly from review text by modeling uncertainty-related linguistic cues including hedging expressions and expressions of conviction within a unified attention-based architecture. We further evaluate strong baseline approaches based on Bidirectional Encoder Representations from Transformers (BERT) and domain-specific SciBERT representations. Our experimental results demonstrate that the proposed attention-based model, grounded in SciBERT representations, achieves encouraging performance compared to strong baselines across multiple evaluation metrics. Importantly, we do not treat hedging as a direct proxy for epistemic uncertainty; instead, it is modeled as a probabilistic linguistic cue whose contribution to confidence estimation is learned contextually through supervised training.

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e-profits : a business-aligned evaluation metric for profit-sensitive customer churn prediction

Información de publicación: Manzoor, Awais; Qureshi, M. Atif; Kidney, Etain; Longo, Luca.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Retention campaigns in customer relationship management often rely on churn prediction models evaluated using traditional metrics such as AUC and F1-score. However, these metrics fail to reflect financial outcomes and may mislead strategic decisions. We introduce *e-Profits*, a novel business-aligned evaluation metric that quantifies model performance based on customer lifetime value, retention probability, and intervention costs. Unlike existing profit-based metrics such as Expected Maximum Profit, which assume fixed population-level parameters, *e-Profits* uses Kaplan–Meier survival analysis to estimate tenure-conditioned (customer-level) one-period retention probabilities and supports granular, per-customer profit evaluation. We benchmark six classifiers across two telecom datasets (IBM Telco and Maven Telecom) and demonstrate that *e-Profits* reshapes model rankings compared to traditional metrics, revealing financial advantages in models previously overlooked by AUC or F1-score. The metric also enables segment-level insight into which models maximise return on investment for high-value customers. *e-Profits* provides a transparent, customer-level evaluation framework that bridges predictive modelling and profit-driven decision-making in operational churn management. All source code is available at: https://github.com/Awaismanzoor/eprofits.

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2L-DQM: a two-level multidimensional quality model for assessing social media data

Información de publicación: Reda, Oumaima; Zellou, Ahmed.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Social media platforms are commonly used to communicate and generate information. In many cases, this information is uncontrolled, which makes it difficult to use and analyze. Although there exist studies focused on data assessment and validation, most of them are limited to specific scenarios. Thus, a more general and flexible model is needed. In previous work, we presented a quality model: FULMQA based on 8 metrics, including credibility, timeliness, popularity, relevancy, accessibility, reliability, presentation and accuracy. In this paper, we propose an extension of FULMQA to provide a robust model to perform various quality analysis of posts on social media based on two levels: content and user, by integrating other different measures and considering various features. The metrics used in 2 L-DQM model were selected with care to make sure that all factors are considered when assessing the quality of the data. While many studies focus on particular and limited features, such as follower counts or engagement rates, etc., our approach takes into account the entire set of relevant indicators. We hope to provide a comprehensive view of social media activity by combining quantitative and qualitative data. This inclusive approach enables us to capture the multidimensional nature of social media interactions. As

a result, our research aims to improve and encourage a better comprehension of data across various social media platforms. With the usage of this model, users will be able to obtain a thorough understanding of the caliber of the data they come across on social media, assisting them in making defensible selections. Then, we performed extensive experiments to evaluate the effectiveness of our model using a Twitter dataset containing 30000 tweets with all the needed features for user and content level. The findings indicate that the model can successfully evaluate all tweets with high performance.

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Correlated bivariate time series forecasting using long short-term memory network: an AutoML approach

Información de publicación: Su, Ying; Wang, Morgan C.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Long-term time series forecasting plays a pivotal but challenging role in automated machine learning (AutoML) research. Forecasting correlated bivariate time series involves modeling and analyzing the dependencies between two historical time series. Temporal correlation, inter-series correlation and long-term dependency are among the major challenges in bivariate time series prediction. Currently, statistical learning-based forecasting models are primarily developed by professionals and requires significant manual effort, including model design, feature engineering, hyperparameter tuning, and model construction. The proposed AutoML approach is designed to assist non-professionals in achieving effective time series forecasts, particularly for applications in finance and economics. Within the AutoML framework, this study proposes the use of Long Short-Term Memory (LSTM) networks to forecast both short- and long-term-correlated bivariate time series, demonstrating superior performance compared with the traditional statistical methods, such as Vector Autoregressive (VAR) and Autoregressive Integrated Moving Average (ARIMA). To streamline AutoML process, the pre-specified hyperparameters are introduced to significantly reduce the need for extensive tuning. A comprehensive analysis of LSTM within AutoML framework is also conducted to evaluate its advantages and limitations. Extensive experiments conducted on both univariate and bivariate time series with varying degrees of correlation show that the proposed method effectively improves prediction accuracy and captures inter-series dependencies.

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Robustness under noise: assessing the impact of perturbed key attributes on machine learning models

Información de publicación: Padró-Ferragut, Cristina; Martínez-Plumed, Fernando; Ramírez-Quintana, María José.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Machine learning models often struggle when confronted with the complexities of real-world data, such as noise and incomplete information, which pose significant challenges for practical applications. This paper presents a systematic framework for assessing the robustness of machine learning models against these adversities. In particular, we investigate how noise affecting the most critical attribute of a dataset impacts on model predictions. Our methodology uses a variety of datasets, a wide range of machine learning models, and a customizable noise induction function, allowing for comprehensive evaluations tailored to specific application needs. Through an extensive experimental analysis, we identify the vulnerabilities of models when their critical attributes are perturbed. In addition, by applying hierarchical clustering, we categorise machine learning families based on their robustness, establishing a taxonomy that highlights the susceptibility of different models to noise. Our findings provide important insights into the interplay between model performance and noise levels and improve the understanding of model behaviour under adverse conditions.

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Human activity recognition (HAR): techniques, applications, and future directions

Información de publicación: Rani, Manjot; Kumar, Munish.

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RESUMEN (ENGLISH)

Human Activity Recognition (HAR) involves the development of intelligent methods to classify and recognize human actions and behaviour in real-world environments by analyzing data acquired from various sensors and sources, extracting meaningful patterns, and making informed decisions about the identified activities. This discipline focuses on training models to understand and interpret human movements using specialized sensors and cameras capable of detecting actions such as walking, sitting, or gesturing. The ultimate aim of HAR is to enhance technology's ability to comprehend human behaviour, contributing to the development of smarter, more responsive devices for domestic applications and ensuring greater safety. This article provides a concise overview of the significance and purpose of HAR, highlighting its applications across diverse domains such as healthcare, sports, and surveillance. The primary objective of this paper is to present a comprehensive review of HAR systems, offering valuable insights to guide new researchers, inform future studies, and deepen the overall understanding of this evolving field.

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An LLM-driven context-aware recommendation system integrating NLP for enhanced social media

personalization

Información de publicación: Yang, Qiwei; Sun, Wei; Habibi, Mostafa; Albaijan, Ibrahim.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The rapid proliferation of social media and location-based services has amplified the demand for intelligent, personalized point-of-interest (POI) recommendations. This paper introduces an advanced AI-driven framework that integrates large language models (LLMs) with deep sequential modeling to enhance sentiment-aware POI recommendation systems. By leveraging cutting-edge neural architectures, including long short-term memory (LSTM) networks and bidirectional encoder representations from transformers (BERTs), our approach effectively captures user interactions, evolving preferences, and contextual factors. A core innovation of our model lies in the application of LLM-powered sentiment analysis on user-generated reviews, extracting nuanced sentiments, implicit feedback, and fine-grained user intent. This enriched understanding enables the dynamic refinement of recommendations, ensuring relevance, diversity, and personalization. Additionally, our framework incorporates temporal dependencies, spatial context, and demographic factors, further enhancing the adaptability and accuracy of recommendations. We validate our approach through extensive experiments on two real-world location-based social network datasets, demonstrating substantial performance improvements over traditional recommendation models. Our findings indicate significant gains in precision, recall, F1-score, and user satisfaction, underscoring the impact of sentiment-aware deep learning methodologies in POI recommendation systems. This research provides valuable insights into the evolving landscape of AI-powered personalization and context-aware recommender systems.

ENLACES

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Predicting lab-grown diamond prices: a comparative view

Información de publicación: Cardoso, Margarida G. M. S; Chambel, Luís.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

We predict lab-grown diamonds' unit prices based on the traditional 4 Cs—Carat (weight), Colour, Clarity and Cut (shape). For comparative purposes, natural diamond prices are also analysed. The data used originated from online diamond retailers. Supervised learning techniques were primarily selected for their interpretability; however, Random Forests were also included due to their strong performance potential, as highlighted in the literature. The Cubist rule-based algorithm achieved the highest predictive performance on lab-grown diamonds data, while on natural diamonds, it ranked second, following Random Forests. Additional insights were provided by the alternative methods used, including Linear Regression, K-Nearest Neighbours, Regression Trees, and Bayesian Networks. In general, unit prices of lab-grown diamonds have proven much more difficult to predict than those of natural diamonds, where the relationship between the key physical attributes (4 Cs) and prices is more evident. Local interpretability was also explored through two queries, one referring to a good-quality diamond with a standard unit weight (1 carat) and another to a larger (5 carat) high-quality diamond. The

Expert's understanding of these queries provided meaningful contributions regarding the formation of diamonds' prices. The findings offer valuable insights into diamonds' price formation and can enlighten consumers and sellers about this constantly evolving market of lab-grown diamonds.

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Constructing ESG-driven portfolios using genetic algorithms based on LSTM-predicted data

Información de publicación: Martínez-Barbero, Xavier; Cervelló-Royo, Roberto; Jordán, Jaume; Ribal, Javier.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Environmental, social, and governance considerations are becoming more relevant for investors, suggesting the need to include ESG metrics in optimizing portfolios to comply with investor requirements. Genetic algorithms offer the flexibility, adaptability, and effectiveness needed to deal with complex multi-objective problems with nonlinear constraints that traditional deterministic methods could not resolve. This paper aims to construct ESG-responsible portfolios that are able to replicate the index's returns and maintain low volatility. We propose a hybrid approach that combines long short-term memory (LSTM) neural networks to predict closing prices and genetic algorithms to solve a multi-objective problem that minimizes tracking error, ESG scores, and volatility. Our analysis uses LSTM-predicted data of the stocks that compose the IBEX 35 and EURO STOXX 50 during 2021 and the first half of 2022. The results show that optimized portfolios present low ESG risk; the returns follow the index closely and have small volatility. This allows investors to replicate the index performance while enhancing the ESG score of the portfolios. Our study addresses the limitations of traditional ESG portfolio optimization by explicitly incorporating predicted returns from long short-term memory networks and leveraging genetic algorithms (GAs) enhanced by NSGA-III.

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Approaches for modelling the term-structure of default risk under IFRS 9: a tutorial using discrete-time survival analysis

Información de publicación: Botha, Arno; Verster, Tanja.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Under the International Financial Reporting Standards (IFRS) 9, credit losses ought to be recognised timeously and accurately. This requirement belies a certain degree of dynamicity when estimating the constituent parts of a credit loss event, most notably the probability of default (PD). It is notoriously difficult to produce such PD estimates at every point of loan life that are adequately dynamic and accurate, especially when considering the ever-changing macroeconomic background. In rendering these lifetime PD estimates, the choice of modelling technique plays an important role, which is why we first review a few classes of techniques, including the merits and limitations of each. Our main contribution, however, is the development of an in-depth and data-driven tutorial using a particular class of techniques called discrete-time survival analysis. This tutorial is accompanied by a diverse set of reusable diagnostic measures for evaluating various aspects of a survival model and the underlying data. A comprehensive R-based codebase is further contributed. We believe that our work can help cultivate common modelling practices under IFRS 9 and should be valuable to practitioners, model validators, and regulators alike.

ENLACES

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Applying metaheuristic optimization for insider-threat detection using natural language processing

Información de publicación: Zivkovic, Tamara; Zivkovic, Miodrag; Jovanovic, Luka; Simic, Vladimir; Radomirovic, Branislav; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Maintaining trust among employees, employers, and institutions is fundamental to business and research, yet the rise of always-online digital systems and expanding workforces poses new risks to operational integrity. Fluctuating work environments, evolving motivations, and gaps in training can leave organizations vulnerable to insider threats, including inadvertent data leaks and intentional exfiltration. In this study, an investigation of natural language processing (NLP) methods applied to HTTP activity logs to identify potential insider threats through behavioral patterns is conducted. Two experiments were conducted using TF-IDF and Word2Vec text representations, each combined with an XGBoost classifier whose hyperparameters were optimized using a newly proposed iteration stagnation-aware variable neighborhood search (ISAVNS) metaheuristic. The ISAVNS introduces a stagnation-detection mechanism that enables adaptive recovery during optimization, improving exploration and convergence stability. Evaluation on publicly available insider-threat datasets confirmed the high effectiveness of the proposed framework, with the TF-IDF-based model reaching an accuracy of 97.63% and the Word2Vec-based counterpart attaining 97.71%.

ENLACES

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Dynamic meta-learning acquisition function method for Bayesian optimization with early stopping criteria for hyperparameter optimization

Información de publicación: Seifi, Farshad; Niaki, Seyed Taghi Akhavan.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Practical hyperparameter tuning is vital for maximizing the performance of machine learning algorithms, as well-tuned hyperparameters are central to advancements in the field. However, the high computational demands of exploring hyperparameters in complex models and large datasets pose significant limitations, often making the tuning process inefficient. Moreover, with the rise of online machine learning applications, the need for rapid response generation has intensified, making algorithm acceleration and data efficiency critical priorities. Meta-learning, or “learning to learn,” offers a promising solution to these challenges by enabling rapid adaptation to new tasks through meta-knowledge from various related tasks. This study proposes a hyperparameter optimization method that combines Bayesian optimization and meta-learning. Our approach leverages prior task knowledge, using similarity measurements derived from meta-features to suggest initial hyperparameters for warm-starting. As the optimization process unfolds, the algorithm dynamically alters similarity measurements to model evaluation techniques, facilitating the generation of effective early responses and continually improving outcomes. With additional evaluations and the development of the surrogate model, the algorithm shifts toward more data-driven Bayesian modeling on the target task, enhancing the quality of the final responses. Moreover, to optimize resource use, the proposed method incorporates a Bayesian learning-curve early-stopping rule that limits resource expenditure on low-potential configurations while avoiding optimistic bias via censoring. Experimental results show that the proposed method delivers, on average, 29.99% better initial performance compared to non-transfer approaches. The final results are also improved, achieving an average accuracy gain of 4.17% over these competitors. Against meta-learning-based methods, the algorithm provides an average improvement of 15.5% in the initial solution and 5.22% in the final solution.

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From headlines to databases: leveraging LLMs for structured disaster event extraction

Información de publicación: Puime Pedra, Matheus; Elkady, Sahar; Villar-Rosety, Fernando M; Colmenero, Alberto; Labaka, Leire; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Disasters, especially those driven by climate change, are putting growing pressure on cities, threatening both their economic stability and social fabric. To build more resilient communities, we need to develop mitigation and preparedness plans. The effectiveness of these plans highly depends on understanding and accessing accurate and comprehensive disaster data. While databases like the Emergency Events Database (EM-DAT) provide useful information, they often exclude smaller, local disasters that do not meet their reporting thresholds. To address this gap, this paper presents a flexible workflow that leverages a large language model (LLM)-driven web scraping tool to collect data on underreported disaster events systematically. The workflow was tested through a case study focused on flood-related incidents in Granada (Spain), demonstrating its ability to capture relevant information. The results indicate strong potential for supporting more

granular disaster damage assessments, with the system achieving an F-score of 76% in identifying requested information. This approach gives communities and policymakers a more complete picture of disaster risks, helping them make better decisions about preparedness and mitigation efforts.

ENLACES

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Analyzing the impact of LLMs in Data Science lifecycle: a systematic mapping study

Información de publicación: Chintakunta, Sai Sanjna; Nascimento, Nathalia; Guimaraes, Everton.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In recent years, Large Language Models (LLMs) have emerged as transformative tools across numerous domains, impacting how professionals approach complex analytical tasks. This systematic mapping study comprehensively examines the application of LLMs throughout the data science lifecycle. By analyzing relevant papers from Scopus and IEEE databases, we identify and categorize the types of LLMs being applied, the specific stages and tasks of the data science process they address, and the methodological approaches used for their evaluation. Our analysis includes a detailed examination of evaluation metrics employed across studies and systematically documents both positive contributions and limitations of LLMs when applied to data science workflows. This mapping provides researchers and practitioners with a structured understanding of the current landscape, highlighting trends, gaps, and opportunities for future research in this rapidly evolving intersection of LLMs and data science.

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ConsistentPeer: reviewers through GraphRAG-driven counterfactuals to measure consistency in peer review

Información de publicación: Bharti, Prabhat Kumar; Panchal, Mihir; Dalal, Viral.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Peer review is the primary mechanism for ensuring the quality of scientific publications. In many conferences, reviewers are required to self-annotate their confidence and provide a recommendation (rating) when evaluating a paper. In cases where the number of submissions is extensive, editors or area chairs rely on these self-rated confidence scores recommendation (rating) to resolve conflicting reviews and borderline cases. However, reviewers may self-assess their confidence in ways that do not align with the content and reasoning presented in their review text, leading to inconsistencies. As a result, these self-assessments may be miscalibrated and fail to accurately reflect the reviewer's conviction regarding the paper's merit.

Such discrepancies arise when reviewers are uncertain about their evaluations or when their judgments lack sufficient supporting evidence. To address this issue, we introduce ConsistentPeer, a novel knowledge graph-based framework (GraphRAG) designed to systematically assess the consistency between textual review content and assigned scores at multiple granular levels, including word, sentence, and aspect-based analysis. Our approach leverages peer review comments from deep learning and natural language processing conferences to construct an enriched representation of review consistency. This work provides a structured framework to identify and mitigate discrepancies in reviewer self-rated scores, ensuring more justifiable and consistent reviews. By doing so, it enhances the reliability and transparency of the peer review process, particularly in borderline cases.

ENLACES

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Improving medical data clustering based on combining density peaks clustering and secretary bird optimization algorithm

Información de publicación: Mahmood, Shaimaa Waleed; Algamal, Zakariya Yahya.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Density peak clustering (DPC) is widely used for identifying clusters in complex datasets but heavily depends on the selection of the cut-off distance parameters, which is often chosen subjectively and can significantly impact clustering quality. To address this limitation, we propose a hybrid algorithm that integrates the secretary bird optimization algorithm (SBOA), a bio-inspired metaheuristic modeled on the hunting behavior of secretary birds, with DPC. The primary contribution of this approach is the automatic optimization of the cutoff distance parameter by efficiently exploring the parameter space and avoiding local optima. This enhancement leads to more accurate and stable cluster identification, particularly in complex medical datasets characterized by high dimensionality and noise. Comparative experiments on benchmark medical datasets against established clustering methods demonstrate that the proposed SBOA-DPC algorithm consistently outperforms alternatives in clustering accuracy and robustness, as measured by metrics such as the adjusted Rand index and adjusted mutual information. These results highlight the effectiveness of bio-inspired optimization in improving parameter selection for clustering algorithms, offering a scalable and reliable tool for medical data analysis and other applications requiring precise cluster detection.

ENLACES

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Complex-order Bat-inspired algorithm applied to neural network optimization for medical datasets

Información de publicación: Esfandiari, Ahmad; Abedi Pahnehkolaei, Seyed Mehdi; Nasiri, Niki.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The Bat-inspired algorithm (BA) is recognized as a prominent member of swarm intelligence algorithms, demonstrating significant potential in addressing global optimization challenges. This algorithm is distinguished by its unique feature of incorporating a local search mechanism, which enhances its convergence speed. However, an overemphasis on exploitation during the initial iterations can result in premature convergence, causing the algorithm to become trapped at a local optimum. To mitigate this limitation and enhance the exploration capabilities, this paper introduces an enhanced version of BA, termed complex-order BA (COBA). The evolution of position and velocity adaption laws in COBA, utilizing the concepts of complex-order derivative and the conjugate-order differential, broadens the memory associated with the prior behaviors of artificial bats and governs the algorithm's convergence. First, we conducted a sensitivity analysis to identify the suitable parameters for the algorithm. Then, the performance of COBA is assessed using the well-known test functions derived from CEC 2017. Furthermore, the performance of the proposed algorithm is investigated for the weight optimization of feedforward neural networks and compared with state-of-the-art metaheuristic algorithms in the literature. COBA was also evaluated on three engineering design problems and consistently outperformed competing algorithms. It achieved superior results on 11 of 14 benchmark functions and showed strong performance in optimizing neural network weights, attaining the highest classification accuracy across all tested datasets. These findings highlight COBA's reliability and effectiveness compared to state-of-the-art methods.

ENLACES

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The generation of power grid anomalous events based on domain-adversarial neural networks

Información de publicación: Qiu, Yuyin; Qian, Jianguo; You, Jiachuan; Lu, Min; Zhang, Xiaonuo; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This paper tackles the challenges of generating and managing power grid anomalies caused by operational variability and unexpected disruptions. Traditional models struggle with adaptability across domains, especially without labeled data or when operational conditions deviate from historical trends. To improve grid stability and reliability, this research introduces key innovations for anomaly generation and simulation under diverse conditions. First, a risk-based methodology is introduced to identify critical transmission lines, integrating a hidden failure model and system-level risk assessments. This method uniquely combines the probability of line failures with their system-wide impact, accurately identifying critical lines during cascading failures. Second, the augmented domain-adversarial network model is proposed, utilizing domain-adversarial learning to address power flow forecasting challenges across varying operational environments. By leveraging graph neural networks for realistic power flow simulations and graph attention networks for feature transformation, the model significantly improves prediction accuracy in scenarios with limited labeled data. Lastly, the integration of the maximum mean discrepancy criterion within a reproducing kernel Hilbert space minimizes feature distribution discrepancies between domains, enhancing the model's adaptability and resilience to dynamic changes in power systems. This research facilitates the generation and simulation of power grid anomalies, enabling timely detection of potential issues. By creating realistic anomalous events, grid operators can better test and refine detection systems,

improving fault identification and response in complex environments. This approach strengthens grid stability and reliability, supporting the development of more resilient power systems.

ENLACES

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DCMatchBoosted - improving deep clustering by architecture recommendation

Información de publicación: Aljoud, Mamdouh; Marques Tavares, Gabriel; Leiber, Collin; Seidl, Thomas.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Deep clustering algorithms such as deep embedded clustering (DEC), deep K-means (DKM), and deep clustering network (DCN) are highly sensitive to the architecture of the underlying neural network, heavily influencing clustering quality. Although neural architecture search (NAS) methods aim at properly configuring deep neural networks, traditional NAS approaches are unsuitable in this context due to the absence of labels. We propose **DCMatchBoosted**, an extension of our previous framework (DCMatch), which leverages dataset characterization and a gradient boosting surrogate to recommend effective autoencoder architectures for deep clustering. Our method combines high-level semantic embeddings from CLIP with complementary statistical descriptors, extracted from a small subset of randomly sampled images, to build a compact representation of each dataset. These dataset features are paired with architecture metadata and used to train an XGBoost model that predicts clustering performance. In extensive experiments on 20 image datasets and three clustering algorithms (DEC, DKM, DCN), DCMatchBoosted consistently outperforms the default configurations, achieving statistically significant improvements in clustering accuracy on the majority of datasets. We make our code available here: Code

ENLACES

Link to OCLC WorldCat

Interpretable rules for online failure prediction: a case study on metro do porto datasets

Información de publicación: Jakobs, Matthias; Veloso, Bruno; Gama, João.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Predictive maintenance applications have increasingly been approached with deep learning techniques in recent years due to their high predictive performance. However, as in other real-world application scenarios, the need for explainability is often stated but not sufficiently addressed, which can limit adoption in practice. In this study, we will focus on predicting failures of trains operating in Porto, Portugal. While recent works have found high-performing deep neural network architectures that feature a parallel explainability pipeline, we find that the generated explanations can be hard to

comprehend in practice due to their low support over the failure range. In this work, we propose a novel online rule-learning approach that is able to generate simple rules that cover the entirety of the detected failures. We evaluate our method against AMRules, a state-of-the-art online rule-learning approach, on two datasets gathered from trains operated by Metro do Porto. Our experiments show that our approach consistently generates rules with very high support that are simultaneously short and interpretable.

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Nalvakku: fact-based counter narratives for Tamil hate speech

Información de publicación: Ravikumar, Narendran; Kumaresan, Prasanna Kumar; Rajiakodi, Saranya; CN, Subalalitha; Chakravarthi, Bharathi Raja.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Warning: This paper contains insulting statements that may cause discomfort for readers. The rapid proliferation of digital platforms has intensified online hate speech, especially in low-resource languages such as Tamil, where automated moderation techniques remain underdeveloped. This paper presents a three-stage methodology for generating counter narratives in Tamil. First, a seed dataset of 220 hate speech counter narrative (HS-CN) pairs is expanded to 5,000 through a human-in-the-loop Author Reviewer framework with expert validation. Second, a fact-based retrieval augmented generation (RAG) system is employed to incorporate external knowledge to enhance factual accuracy and persuasiveness. Finally, the human post-edited dataset is integrated to the RAG system as a curated knowledge base yielding a Fact-RAG system with stronger factual grounding and cultural appropriateness. Assessment through intrinsic indicators and LLM-based evaluations indicates that our methodology generates counter-narratives that are varied, credible, and contextually relevant. These findings underscore the effectiveness of integrating human supervision, factual validation, and selected examples for counter-narrative development in low-resource settings. **GitHub:**https://github.com/Bharathi-AI-for-Social-Good/Fact-RAG-BasedCN-Ta

ENLACES

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TA-STGCN: trend-aware attention spatio-temporal graph convolutional network for traffic flow prediction

Información de publicación: Chen, Guihui; Chen, Lihui; Li, Zhongbing; Wei, Yuli; He, Zhili; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Traffic flow prediction faces significant challenges in effectively capturing dynamic traffic patterns and their time-varying features. Although existing deep learning models have achieved some results, accurate traffic prediction still remains a complex task. Existing methods are often limited by their focus on discrete traffic data, ignoring key information such as continuous trend changes, which can hinder the prediction accuracy. To address these issues, this study proposes a trend-aware attention Spatio-temporal Graph Convolutional Network (TA-STGCN) for traffic flow prediction. It designs the trend-aware attention mechanism to effectively capture local traffic flow variations and develops a dual-channel attention module to comprehensively capture the temporal correlation of traffic flow. In addition, trend information is introduced to optimize the learning of dynamic spatio-temporal graphs, while a multi-graph convolutional network is used to encode both the dynamic spatial information and trend data of traffic nodes, thus modeling complex spatio-temporal dependencies. Evaluation on four real-world datasets confirms that our proposed TA-STGCN achieves superior prediction accuracy compared to current baseline methods.

ENLACES

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Categorical variable encoding methods for tabular data: a benchmarking study

Información de publicación: Clerici, Federico; Nobani, Navid.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Machine learning models often require numerical inputs, making the encoding of categorical features a critical step in the data preprocessing pipeline. A wide range of encoding methods, such as the commonly used one-hot encoding, are available, but they may not always be optimal due to increased dimensionality and a lack of sensitivity to the inherent relationships between categories. This paper presents a comprehensive evaluation of 26 categorical encoding techniques, benchmarked across 13 real-world datasets and 7 different machine learning algorithms. Our study categorizes these methods based on predictive task type, model performance, and computational efficiency, offering a taxonomy for selecting encoders. In addition, we illustrate how Safe AI metrics can be applied to encoding pipelines, showing that they provide complementary insights into model robustness and fairness. Finally, we provide a Python tool called EncodeHero that enables researchers and practitioners to (1) extend the results by augmenting the benchmark with their own data and (2) choose the best encoding methodology based on their data and technical constraints.

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Global natural gas trade forecast via hybrid machine learning

Información de publicación: Yan, Yu; Lu, Gang; Yan, Xiaoqing; Xia, Peng; Deng, Song; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Accurate global natural gas trade forecast is crucial for a country to optimize and adjust its energy strategy to ensure energy security. Since various complex and uncertain factors influence global natural gas trade, traditional forecasting methods based on expert knowledge and statistical methods make it difficult to achieve accurate forecasts. Recently, approaches based on machine learning (ML) have been adopted to model such complex and uncertain factors based on historical data associated with global natural gas trade. However, global natural gas trade has three special characteristics: (1) network characteristic, i.e., the trade network relationships between various countries, (2) dynamic characteristic, i.e., the trade relationships dynamically change over time, and (3) complex-factor characteristic, i.e., the trade relationships are influenced by many complex factors. Unfortunately, none of the existing ML models can comprehensively capture these three characteristics for forecasting. To fill this gap, this paper proposes a novel hybrid ML model by integrating three ML models of Graph Attention Networks (GAT), Gated Recurrent Unit (GRU), and Transformer, which is termed GAT-GT. The main idea of GAT-GT is threefold: (1) GAT is employed to learn the latent features of the annual trade network among various countries year by year, (2) the learned time-sequential latent features are input into GRU to model the dynamic trade relationships, and (3) Transformer is exploited to combine various complex factors with the learned latent features for forecasting. As such, GAT-GT enjoys all the merits of GAT, GRU, and Transformer, making it able to comprehensively capture these three characteristics of the global natural gas trade. In the experiments, the global natural gas trade data from 2000 to 2023 were collected to benchmark our empirical studies. The results demonstrate that our GAT-GT model significantly outperforms eight advanced ML competitors in forecasting the global natural gas trade.

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CNN-based deep learning approach for multiclass prediction of fin-tech startups

Información de publicación: Rokhashmah, Morteza; Ekinci, Cumhur.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Fin-tech startups, while pivotal to financial innovation, face exceptionally high failure rates, underscoring the need for advanced predictive tools to guide investment and policy decisions. This study introduces a deep learning framework based on 1D Convolutional Neural Networks (CNNs) to classify fintech startup outcomes into four categories: Initial Public Offering (IPO), Merger & Acquisition (M&A), Closure, and Unicorn status using global Crunchbase data. CNN extracts local and hierarchical patterns from structured scalar attributes such as team composition, funding velocity, investor diversity, and company age. To address class imbalance, we adopt ADASYN oversampling. Compared to traditional models, XGBoost achieves 92.9% accuracy and 91.9% macro F1, while the CNN model achieves 90.8% accuracy and 89.6% macro F1, demonstrating competitive performance and improved sensitivity to rare outcomes such as IPOs and Unicorns. The results confirm tree-based dominance on structured data, although CNNs provide a viable deep-learning alternative for multiclass startup outcome prediction.

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Attention-enhanced BiLSTM for causal sentiment mining in noisy social-media streams

Información de publicación: Mihoubi, Miloud; Zerkouk, Meriem; Chikhaoui, Belkacem.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Short, noisy social-media messages make sentiment analysis particularly challenging: slang, emojis, and misspellings not only reduce predictive accuracy but also obscure why a model outputs a given polarity. This paper introduces a lightweight hybrid model that combines a convolutional block, a bidirectional long short-term memory network (BiLSTM), and a single multi-head self-attention layer with a novel noise-invariant contrastive head (NICH) that normalizes noisy tokens before encoding. The model is trained in two phases (frozen then fine-tuned word embeddings) and evaluated on two contrasting benchmarks: Sentiment140 (1.6M tweets) and IMDb (50k movie reviews). Across both datasets, our approach outperforms classical baselines (logistic regression, support vector machine, random forest) and an Attention-BiLSTM reference model, while remaining competitive with a compact Transformer-based baseline (DistilBERT), achieving 90.63% accuracy and 0.9079 F1 on Sentiment140 and 91.90% accuracy and 0.9203 F1 on IMDb. Attention scores and a perturbation-based metric show that the model consistently focuses on sentiment-bearing tokens, providing token-level explanations that are useful for real-time monitoring of customer feedback and brand perception in noisy social-media streams.

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Bayesian causal forests with covariate balancing propensity score: a novel approach for heterogeneous treatment effects estimation

Información de publicación: Rafiullah; Wang, Hong.

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RESUMEN (ENGLISH)

Bayesian causal forests (BCF) are a powerful method for estimating heterogeneous treatment effects (HTE) in the presence of confounding variables. However, the accuracy of BCF critically depends on the correct estimation of propensity scores (PS); misspecified PS can lead to biased estimates of the treatment effect. To address this limitation, we propose BCF with covariate balancing propensity scores (BCF-CBPS), which integrates covariate balancing directly into the PS estimation step. By replacing conventional PS with CBPS, our approach improves covariate balance, enabling BCF to more accurately

estimate true treatment effect heterogeneity. Extensive Monte Carlo simulations and real-world applications demonstrate that BCF-CBPS achieves higher precision and robustness compared to standard BCF and other machine learning methods for HTE estimation. These results highlight the effectiveness of combining covariate balancing with the BCF model for reliable estimation of treatment effects in complex observational studies.

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Reinforcement learning for energy management in fuel cell hybrid electric vehicles: an up-to-date review with topology-centric insights

Información de publicación: Sahbani, Abdelaziz; Ben Saad, Kamel.

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RESUMEN (ENGLISH)

The appropriate design of an energy management system (EMS) is crucial in fuel cell hybrid electric vehicles (FCHEVs) as it must ensure an efficient coordination of the multiple energy sources including the fuel cell (FC), battery, and super-capacitor (SC). This paper presents an up-to-date and extensive review of EMS strategies, focusing on RL-based methodologies and their application on new powertrain architectures. RL is identified as a promising technology that could be used to implement intelligent and adaptive control solutions that would also be energy efficient for FCHEVs. The review focuses on four main aspects: (1) powertrain configurations adopted for FCHEVs, (2) reinforcement learning techniques, parameters, and performance measures, (3) the use of multi-objective optimization for EMS design, and (4) datasets and simulation cycles applied in the studies being reviewed. Special focus is placed on the ability of the RL framework to learn dealing with conflicting objectives. These insights aim to guide researchers in advancing next-generation EMS for FCHEVs.

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Systematic literature review on sentiment analysis using transformers

Información de publicación: Mostafavi, Sepehr; Yahyavi, Yeganeh; Ravanmehr, Reza.

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RESUMEN (ENGLISH)

With the rapid expansion and growing complexity of social networks, developing effective approaches for extracting meaningful knowledge from the vast volumes of user-generated data has become an essential challenge. Among these approaches, sentiment extraction and analysis- one of the core tasks in natural language processing (NLP)—plays a critical

role in uncovering users' attitudes, opinions, and emotions. Insights gained from such analysis can inform strategic decision-making across a wide range of domains, including marketing, public policy, healthcare, and service management. Over the past decade, sentiment analysis has attracted increasing attention within interdisciplinary research, with many studies leveraging artificial intelligence-based techniques. In particular, transformer architectures have emerged as leading methods in NLP, primarily due to their capacity to model long-range textual dependencies and to generate richer, more precise semantic representations. This paper presents a systematic literature review of research on transformer-based sentiment analysis. Specifically, we categorize and analyze different models, widely adopted benchmark datasets, evaluation metrics, as well as the challenges and diverse applications associated with these methods. A qualitative assessment of the reviewed studies, based on established systematic review standards, highlights the progress and maturity of the field from technical, linguistic, and applied perspectives. Furthermore, we analyze current research trends and identify promising directions for future work. The findings of this review underscore both the substantial growth in research output and the increasing significance of transformer-based sentiment analysis, reflecting its dynamism, innovation, and impact on interdisciplinary studies.

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D3A-TS: denoising-driven data augmentation in time series

Información de publicación: Solís-Martín, David; Galán-Páez, Juan; Borrego-Díaz, Joaquín.

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RESUMEN (ENGLISH)

It has been demonstrated that the amount of data is crucial in data-driven machine learning methods. Data are always valuable, but in some tasks, it is almost like gold. This occurs in engineering areas where data are scarce or very expensive to obtain, such as predictive maintenance, where faults are rare. In this context, a mechanism to generate synthetic data can be very useful. While in fields such as computer vision or natural language processing, synthetic data generation has been extensively explored with promising results, in other domains such as time series it has received less attention. Previous works have proposed techniques like geometric transformations, interpolation, and generative models like GANs and VAEs for time series data augmentation. However, the use of denoising models, particularly diffusion probabilistic models (DPMs), remains largely unexplored in this context. This work specifically focuses on studying and analyzing the use of different techniques for data augmentation in time series for classification and regression problems. The proposed D3A-TS methodology involves the use of DPMs, which have recently achieved successful results in the field of image processing, for data augmentation in time series. Additionally, the use of meta-attributes to condition the data augmentation process is investigated. The results highlight the high utility of this methodology in creating synthetic data to train classification and regression models. To assess the results, six different datasets from diverse domains were employed, showcasing versatility in terms of input size and output types. Finally, an extensive ablation study is conducted to further support the obtained outcomes.

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Multi-view subspace clustering via low-rank consensus projection matrix

Información de publicación: Zeng, Qing; Yu, Yue; Zhang, Wei; Fan, Zizhu.

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RESUMEN (ENGLISH)

Multi-view subspace clustering (MVSC) uncovers the intrinsic structure of multi-view datasets by integrating complementary information from diverse views. However, most existing methods neglect the simultaneous integration of consistency and complementarity in multi-view data, leading to degraded clustering performance. We propose a novel multi-view subspace clustering method, termed LRPMSC, which employs a projection matrix to map raw data into a lower-dimensional space. This mapping mitigates noise interference and preserves local structures while facilitating distance measurement between samples. Furthermore, LRPMSC adopts a matrix tri-factorization strategy to extract consistent structures from the representation matrix. The Schatten *p*-norm is imposed on the core matrix as a low-rank constraint, enhancing robustness against high-dimensional noise. Moreover, angular information and fusion mechanisms are incorporated to refine the final similarity matrix. To systematically evaluate LRPMSC, we conduct extensive experiments on nine benchmark datasets and compare its performance against twelve state-of-the-art multi-view clustering methods.

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A novel deep learning approach with feature injection technique for automated gait detection in parkinson's disease using multi-modal data

Información de publicación: Sayem, Faizul Rakib; Ahmed, Mosabber Uddin; Khalil, Hanan; Pedersen, Shona; Ali, Mohamed Sultan Mohamed; et al.

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RESUMEN (ENGLISH)

Parkinson's disease (PD) is a persistent, irreversible neurodegenerative condition, with freezing of gait (FOG) being among its most incapacitating symptoms. PD patients with FOG encounter abrupt and unforeseen difficulties in initiating or maintaining movement, potentially resulting in unexpected falls and injuries. A sensor-driven FOG detection system can facilitate continuous and objective tracking of PD patients experiencing FOG and provide on-the-spot cueing support. Many recent studies have employed advanced deep learning methods to detect FOG by analyzing data from only inertial measurement unit (IMU) sensors. In our study, we have utilized a multimodal dataset that includes a gait accelerometer (ACC), electromyogram (EMG), and electroencephalogram (EEG) for FOG detection. For robust FOG detection task, we have proposed a novel deep learning-based model called Self-FOGNet, which integrates the self-organized operational neuron (Self-ONN) layer to enhance the learning of intricate patterns within the data by introducing non-homogeneity in neural networks. The robust model also promotes feature reuse and efficiently utilizes the hierarchical feature space across

multiple layers, enabling effective classification. Further, we extracted several wavelet features and injected them at the last layer of the model as time domain and frequency domain features. We have used 3 different models for 3 data modalities and used an ensemble learning approach to combine the outputs of each model to get the final improved prediction. As a result of our proposed method, we achieved 98.74% accuracy and 98.68% specificity in the FOG detection task with an ensemble learning approach that outperformed the state-of-the-art approaches.

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A hybrid deep learning and metaheuristic optimization algorithm for accurate and scalable community detection in complex networks

Información de publicación: Pourmahmoud, Solmaz; Gharehchopogh, Farhad Soleimanian; Masdari, Mohammad.

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RESUMEN (ENGLISH)

Community detection is a critical challenge in analyzing complex networks, with traditional methods often struggling with scalability and accuracy in large and intricate systems. This paper proposes a novel hybrid approach, MGO-DE-LA-BP-CDDL, to address these issues. This method adapts multiple techniques, including Mountain Gazelle Optimizer (MGO), Differential Evolution (DE), Learning Automata (LA), and Deep Backpropagation (Deep BP). The framework utilizes MGO-DE-LA for global optimization while using the BP technique for local refinement, ultimately enhancing community detection accuracy. The method incorporates a self-adaptive mechanism that dynamically balances exploration and exploitation, enabling robust optimization and uncovering hidden network structures. Parallelized and distributed computations on CPUs and GPUs further improve scalability and efficiency. The proposed method employs two objective functions to guide the optimization process and integrates stacked unsupervised learning for feature extraction with supervised BP algorithms for refinement. Experimental evaluations on real-world datasets demonstrate that the proposed method significantly outperforms traditional approaches, achieving higher modularity scores and improved community detection accuracy. These results highlight the proposed method's potential to advance network science by providing a scalable and efficient solution for analyzing complex systems.

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Lanxensemble: explainability enabled landslide susceptibility prediction

Información de publicación: Sreelakshmi, S; Chandra, S. S. Vinod.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The identification of landslide risk aids in accurately mapping susceptibility, which is essential for disaster relief operations and sustainable land use planning. Due to inaccessible landslide distributions, typical detection techniques are mostly ineffective because of their considerable workload, subjectivity, and inaccuracy. This research proposes an explainable machine learning strategy for landslide risk identification that addresses these constraints. To improve the identification of different possible landslide types, the method is built on a multistage fusion architecture that integrates knowledge about probable landslide-causing factors. First, an information gain filter-based feature selection method is used to select the most relevant features and train several linear and ensemble classification models using the filtered features. After that, it makes a landslide prediction and explains, respectively. When compared with other machine learning models, this ensemble model provides better performance in landslide prediction tasks. Also, the proposed model uses some effective visualization techniques for accurate explainability tasks.

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Current English–Chinese legal language mutual translation based on the comparison of the three theoretical levels of words, sentences, and articles

Información de publicación: Nan, Yuxiang.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The history of Chinese and foreign legal systems and legal language shows that the long-term development of each legal system, and its language expression system is inseparable from the absorption of the advantages of other legal systems through translation. With the aid of the "translation stream," the modern socialist legal system with Chinese features is progressively reaching the vast nirvana that incorporates the benefits of diverse legal systems. In order to obtain more efficient and intelligent legal language translation, the study conducts comparative analysis at the theoretical level of words, sentences and texts, proposes translation principles and methods with legal characteristics, and develops a multi-level knowledge-based neural machine translation model. The research results show that after training, the MKNMT model only iterates 308 times, and the loss value can reach the minimum; the MKNMT model only iterates 316 times to achieve the target accuracy. In different training sets and translation tasks, the BLEU value of the MKNMT model is the highest. From the overall analysis, the lower the perplexity of the model, the higher the BLEU value. To sum up, the translation performance of the MKNMT model proposed in the study is very good, which successfully raises the machine translation model's translation quality, obtains better application results in the prevalent English–Chinese legal language, and increases the interest of legal readers in reading.

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Deep learning for object detection: state of the art, challenges, and future directions

Información de publicación: Gupta, Neetika; Rohil, Mukesh Kumar.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Starting from the use of deep belief networks as a fast learning algorithm, deep learning techniques have proved to be efficient in number of ways because of their capability of learning self-defined features rather than operating on fed features. The concept has drawn ever-increasing research interest in recent years and has been achieving remarkable performance in learning tasks involving text, sound, or image/video components. Due to its promising performance, there have been many efforts made to evolve deep learning to better suit to various applications that may gain substantial improvement over the state-of-the-art methods. With an increased demand of autonomous vehicles, smart video surveillance systems, facial gesture, human detection applications, etc., fast and accurate object detection systems constitute a very promising area of research and development. Robust object detection requires recognition and classification of every object in an image under unfamiliar environmental conditions such as occlusion and cluttered objects in a confined space. Therefore, localizing and labeling each object in an image by drawing an appropriate bounding box around it makes object detection a challenging task. The article reviews methods which rely on the application of deep learning to object detection and recognition. A cutting-edge review is accomplished on various deep learning architectures and frameworks, summarizing recent progress in the domain. The paper also discusses promising future research directions by elaborating on the limitations of the deep learning techniques applied for detection and recognition of objects.

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A comprehensive survey on multi-behavior recommender systems: extended taxonomy and recent advances

Información de publicación: Kim, Kyungho; Kim, Sunwoo; Lee, Geon; Jung, Jinhong; Shin, Kijung.

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RESUMEN (ENGLISH)

Traditional recommender systems primarily rely on a single type of user-item interaction, such as item purchases or ratings, to predict user preferences. However, in real-world scenarios, users engage in a variety of behaviors, such as clicking on items or adding them to carts, offering richer insights into their interests. Multi-behavior recommender systems leverage these diverse interactions to enhance recommendation quality, and research on this topic has grown rapidly in recent years. This survey provides a timely review of multi-behavior recommender systems, focusing on three key steps: (1) Data Modeling: representing multi-behaviors at the input level, (2) Encoding: transforming these inputs into vector

representations (i.e., embeddings), and (3) Training: optimizing machine-learning models. We systematically categorize existing multi-behavior recommender systems based on the commonalities and differences in their approaches across the above steps. Additionally, we discuss promising future directions for advancing multi-behavior recommender systems.

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Fusion of pre-trained model features with style-aware attention for enhanced content-based recommendation system

Información de publicación: Suvarna, B; Balakrishna, Sivadi.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Recommendation systems are integral to online platforms and e-commerce applications, aiding users in discovering personalized and relevant content or products. Recent advancements have showcased the effectiveness of attention mechanisms in enhancing recommendation system performance by focusing on informative data segments. However, there is a need to retrieve similar products based on the style available in the product image, which is crucial for recommendation systems. Motivated by the need to identify style-based patterns, this study leverages style attention-based deep learning architectures to enhance recommendation system performance. The proposed approach comprises two key tasks: Query Product Classification and Extracting Product Recommendations. The first task introduces a Dual Attention Network, which takes the attention weights jointly obtained from the Channel Style-Based Attention Module (CSAM) and Gated Style-Based Attention Module (GSAM) to categorize the query product. This approach starts with extracting two feature representations for the query product image using Xception and VGG-16. The channel attention is applied to the feature representation derived from the Xception, which is then processed through the Style-Based Attention Module. Similarly, gated attention is applied to the feature representation derived from VGG-16, which also passes through the Style-Based Attention Module, to capture style patterns in the local feature maps. Subsequently, these features are fused and fed into a network to categorize the query product. The cosine similarity is computed to retrieve analogous products, enabling the detection of visually similar items within the product repository. By analyzing style-related features, recommendation systems gain valuable insights into users' underlying preferences and affinities, which may not be overtly articulated. By integrating attention mechanisms and diverse feature representations, the proposed system achieves an accuracy of 92.16%,82%, and 98.27% on the Fashion Product Images, Shoe, and Shoe vs Sandal vs Boot Image datasets, respectively, outperforming numerous existing models.

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A data-driven analysis to predict energetic intelligence

Información de publicación: Baca Ruiz, Luis Gonzaga; Criado-Ramón, David; Serrano-Fernández, María José; Pérez-Moreiras, Elena; Pegalajar Jiménez, María del Carmen.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Energetic Intelligence is a newly defined construct recently validated that offers a thorough understanding of human intelligence by integrating, emotional, spiritual, and physical dimensions. This study applies a data-driven approach to predict Energetic Intelligence. Our research goal lies in the application of machine learning techniques to model and predict Energetic Intelligence using data commonly gathered in organizational contexts. We collected responses through structured surveys and employed a range of supervised learning algorithms to build predictive models. Model performance was evaluated using standard metrics, with the best results reaching an R^2 of 0.73 through optimized and simplified models, which is a promising outcome for a psychologically grounded prediction task. Key predictors included variables such as Flow, Flourishing, and Emotional Vitality, which consistently emerged as relevant features in model training. These findings demonstrate the potential of machine learning to support psychological research and offer practical tools for the assessment and development of Energetic Intelligence in applied settings. Our work points out the value of integrating AI methodologies with psychological theory to enable data-driven insights into human potential and well-being.

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A Kendall's rank correlation coefficient-based decision tree for monotonic classification problem

Información de publicación: Mu, Yashuang; Sun, Yan; Hu, Yi; Wang, Lidong; Guo, Hongyue; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This paper presents a monotonic decision tree framework to tackle ordinal classification challenges by integrating correlation screening, rank mutual information, and monotonicity constraints. The proposed monotonic decision tree introduces a two-stage screening mechanism under the principle of "threshold admission, information first" to select the optimal splitting feature. In the first stage, the Kendall's rank correlation coefficient is utilized to screen for candidate feature set satisfying the monotonicity threshold. In the second stage, we obtain the final splitting feature by combining the Kendall's rank correlation coefficient and rank mutual information. The experimental results demonstrate that the proposed monotonic decision tree is feasible and effective in terms of model compactness, ordinal prediction accuracy, noise robustness, and construction efficiency.

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Meta-path attention with semantic transformer for academic recommendation

Información de publicación: Wang, Wei-Cheng; Huang, Hai-Yin; Wu, Tung-Yang; Hwang, Ming-Jiu; Wang, Jun-Zhe; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In recent years, the rapid growth in scholarly publications and the widespread adoption of digital libraries have intensified the demand for effective paper recommendation systems. Traditional approaches typically rely on extensive historical interaction data or rich contextual information from paper abstracts, but often overlook critical temporal dependencies inherent in user preferences. A particular challenge is the recommendation of newly published papers, which, despite their significance in conveying cutting-edge research findings, suffer from sparse historical data. To address these issues, we propose the Meta-path Attention with Semantic Transformer for Academic Recommendation (MAPSTAR) framework, a novel recommendation model that integrates heterogeneous graph attention with transformer-based meta-path attention mechanisms. MAPSTAR simultaneously models both the temporal sequences of user interactions and the complex correlations among papers and their attributes. Specifically, our approach introduces a Transformer Encoder within the meta-path attention layer, allowing each meta-path embedding to capture global dependencies and dynamically adjust its representation based on contextual interactions with other meta-paths.

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Zoning of soybean production in the state of Mato Grosso, Brazil, from 1990 to 2018, by cluster analysis

Información de publicación: Damian, J. G. R; Piedade, S. M. S; Vale, G. M; Oliveira, V. J. S.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The present research aimed to use the clusters analysis technique to create groups and rank them in the 141 municipalities of the state of Mato Grosso located in Brazil, from 1990 to 2018, for the soybean production variables in thousand tons, value of production in thousand reais (R$) and value of derivatives in thousand reais (R$). The proposed methodology was based on the use of the cluster analysis technique, using the DTW (dynamic time warp) distance together with Ward's hierarchical clustering method, to create the groups, creating three dendrogram structures for each of the variables, and thus the validation of the adequate number of groups occurred by the Mojena method in the dendrogram structures together with Pearson's statistical correlation test followed by the Mantel test. Dendrograms with 5, 5 and 4 groups were constructed and statistically validated and presented in a staggered way for each variable, respectively, and it was found that they show an increasing trend in each of these variables. It was observed that the development of these variables in the state occurs from the central municipalities to the most peripheral ones and bordering the main highways that cross the state. The present research generated an important staggered portrait of variables linked to the soybean production sector in the state of Mato Grosso in this period, and these results can attract both internal and external investors such as China

and other countries interested in this sector.

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Aqnet adaptive mesh attention framework for spatiotemporal air quality prediction in diverse urban environments

Información de publicación: Fusiripong, Prashaya; Surbakti, Herison.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Urban air quality forecasting is vital for managing pollution exposure and protecting public health. This study introduces AQNet, a spatiotemporal deep learning framework that integrates adaptive mesh-based graph construction with attention-driven temporal modeling to enhance prediction accuracy. The model is tested on the Beijing Multi-Site Air Quality Dataset containing hourly pollutant and meteorological records from 12 monitoring stations. AQNet captures spatial dependencies through graph convolution over nonuniform mesh partitions and models temporal dynamics using gated recurrent units. A cross-fusion layer integrates these representations before final prediction. The framework surpasses existing models such as those by Han et al. and Chen et al., achieving a Mean Absolute Error of 2.50, Root Mean Square Error of 3.35, and an R2 of 0.96, with precision, recall, and F1-score above 0.95. The adaptive mesh and attention modules enable efficient scaling across cities with different sensor densities, as confirmed through runtime and GPU profiling across Beijing, Delhi, Bangkok, and Kathmandu datasets. AQNet maintains high-resolution forecasting accuracy with minimal computational cost, confirming its suitability for deployment in diverse urban environments.

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An approach for detecting emerging operational risks from textual data

Información de publicación: Piacenza, Fabio; Greselin, Francesca; Di Vincenzo, Davide; Zitikis, Ričardas.

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RESUMEN (ENGLISH)

Operational risk (OpRisk) constitutes a significant non-financial threat with wide-ranging implications for financial institutions. While traditionally centered on regulatory compliance—encompassing data collection, capital requirement estimation, and the generation of managerial reports—OpRisk functions are increasingly shifting toward proactive strategies designed to anticipate and mitigate potential risk exposures. Artificial Intelligence techniques are playing an

expanding role in this transformation by extracting actionable insights from unstructured data sources. In this study, we investigate the application of text analysis methods, a key component of natural language processing, to OpRisk event descriptions. The contributions of this work are threefold. First, we present a fully integrated workflow that applies text analysis techniques not only to internal event narratives but also to external data sources, including publicly available web content. Second, we introduce a tailored variant of latent Dirichlet allocation for clustering OpRisk events, aimed at uncovering latent patterns and identifying the underlying causes of risk. Third, recognizing the limitations of conventional models in mitigating the impact of future loss events, we explore the enhancement of traditional data sets with alternative sources that enable earlier detection of emerging risks. Specifically, we develop a real-time analysis of relevant content from X (formerly Twitter) to enable continuous monitoring of the evolving risk landscape and to detect early warning signals of novel risk types. This study demonstrates how these diverse methodologies can be cohesively integrated into a unified OpRisk management framework, fostering a more holistic, forward-looking, and adaptive approach to risk mitigation.

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Lazy lossless compression by symbol lookahead invalidation in universal adaptive stream-based entropy coding

Información de publicación: Yamagiwa, Shinichi; Kato, Taiki; Nishikawa, Taku.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This paper focuses on a novel technique on lossless compression mechanism for streaming data based on lookup table. The technique improves its compression ratio and to reduce the execution time, where the compressor encodes a symbol by translating it to an index of the table. In our advanced research outcomes, we have proposed ASE (adaptive stream-based entropy) coding that compresses a unit of data in streaming data by shrinking the table index to the number of bits based on real-time data entropy. The entropy is calculated based on the number of occupied entries in the table. To improve the compression ratio, we should reduce the number of occupied entries in the table during processing streaming data. We further proposed a pipeline-based lossless data compression mechanism called universal ASE coding. It connects multiple ASE coding modules to compress frequent symbol sequence. This mechanism needs a large computation power in software implementation due to the multiple ASE coding modules. In this paper, we propose symbol lookahead invalidation that discards original symbols in the compression pipeline at table hits. The invalidation avoids registration to the table of ASE coding in the compression pipeline, and the number of occupied entries in the table is reduced. This also speeds up the compression process by skipping ASE coding operations. According to experimental evaluation, we confirmed the compression ratio was improved slightly, and 30% of the execution time was improved. This paper describes that the invalidation method is applicable to the conventional compression algorithm and we can expect the similar effects from the technique.

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A systematic literature review on explainable AI for churn prediction, customer segmentation and retention

Información de publicación: Maqbool, Misbah; Qureshi, Ishtiaq Hussain.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Nowadays, the incorporation of XAI—explainable artificial intelligence—into customer analytics has gained major attraction driven by the need for trust, transparency and actionable understanding in predictive modelling. This systematic literature review analyses the application of XAI techniques like SHAP, LIME and also other AI models like machine learning, deep learning, ensemble models and further provides counterfactual explanations in churn prediction, customer segmentation and retention strategies. Using a structured search across important databases like Scopus, IEEE and other digital libraries, 80 studies published between 2015 and 2025 were selected and analysed. This review categorises XAI approaches by industry application, interpretability findings and methodology and emphasises their role in improving model decision-making and transparency. The findings explore that XAI not only enhances stakeholder trust but also enables more ethical and targeted customer interventions. A comparative study has also been performed for XAI techniques for churn prediction, customer segmentation and retention. The limitations in real-time deployment, standardisation and evaluation metrics are discussed in gaps and discussions along with future research directions for XAI models, hybrid models and domain-specific explainability contexts.

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A fully decentralized federated adversarial vision transformer with blockchain and secure aggregation for visual-based intrusion and malware forensics

Información de publicación: Belal, Mohamad Mulham; Saleti, Sumalatha; E, Elakkiya.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This paper presents a fully decentralized federated adversarial vision transformer (ViT) framework for secure, privacy-preserving, and robust image-based malware classification. Unlike conventional federated learning that relies on centralized aggregation and remains vulnerable to privacy breaches and adversarial attacks, the proposed system employs blockchain-based decentralized aggregation integrated with secure multi-party computation. Encrypted local model updates are securely aggregated without a central server, while the blockchain ledger ensures transparency, tamper resistance, and trust. To further enhance security, a zero-knowledge proof-based mechanism validates masked model updates, enabling verifiable aggregation without exposing raw parameters. Clients reconstruct the global model through

decentralized consensus, preventing direct access to others' updates. Adversarial robustness is improved via client-side adversarial ViT training, incorporating projected gradient descent-generated malware images with clean samples, thereby reducing false classifications. Computational efficiency is achieved by leveraging pre-trained ViT variants for resource-constrained environments. Extensive experiments on Malimg, Microsoft BIG 2015, and Malevis datasets demonstrate superior performance, achieving accuracies of 98.30%, 98.93%, and 95.72%, respectively. Compared to centralized and federated adversarial ViTs, as well as state-of-the-art methods (FASe-ViT, FASNet, DM-Mal, Fed-Mal), the proposed framework consistently achieves higher accuracy, precision, recall, and F1-scores, while ensuring privacy, resilience, and decentralized trust.

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Average Volatility Dimensioning (AVD): dimension reduction technique for multivariate time series

Información de publicación: Mallinger, Kevin; Marica, Edina; Schrittwieser, Sebastian; Raubitzek, Sebastian.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Reducing the features while preserving a dataset's inner dynamics is a challenging task. This is particularly evident for nonlinear and time-dependent processes. We propose a novel dimension reduction technique, Average Volatility Dimensioning (AVD), which captures the behavior between features and reduces the multiple dimensions to a univariate signal. This article describes the developed mechanisms and compares this dimension reduction technique against nine state-of-the-art methods. The ability to retain the information of the data is evaluated by using the reduced dimensions for machine learning classification tasks and comparing the phase-space portraits. The validation is performed on eight different datasets featuring nonlinear time series data, covering domains such as movement recognition, fault detection, and environmental monitoring. Overall, the results show that AVD encapsulates the data dynamics best, demonstrating superior performance for the discussed machine learning approach while maintaining an interpretable phase-space trajectory, resisting noise, and remaining computationally efficient.

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An empirical evaluation of clustering processes for early detection of university dropout

Información de publicación: Melchor, Fran; Conejero, José M; Fernández-García, Antonio Jesús; Sánchez-Figueroa, Fernando; Rodríguez-Echeverría, Roberto.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The elevated rates of dropout within academic institutions have prompted the use of Artificial Intelligence (AI) to tackle this issue. These efforts often rely mainly on administrative and academic data, lacking personal information about students. In a previous study, we explored machine learning models to leverage this data and harness their knowledge-extraction capabilities. However, a critical factor, the availability of labeled data, was not addressed. Obtaining these data may be challenging due to their distribution across different systems or the considerable time required to collect them, especially when new degrees are being implemented. The lack of labeled data is a major obstacle for institutions that do not possess them so that they are unable to take advantage of the full potential of AI for their purposes. Clustering algorithms have conventionally been employed to uncover latent patterns within unlabeled data. These unsupervised algorithms may reduce the need for data labeling; nonetheless, it necessitates rigorous validation of the resulting clusters, particularly when dealing with datasets encompassing numerical and categorical attributes. This paper introduces a comparison of various clustering algorithms to discern the most appropriate technique for uncovering the underlying factors contributing to university student attrition, employing unlabeled data. The novelty lies not only in the algorithmic comparison but also in their integration with diverse data preprocessing methodologies, streamlining the selection of the optimal combination including advanced data transformations for the harmonization of numerical and categorical information. It is illustrated through a real-world case utilizing academic data from a Spanish university, providing empirical validation for the proposed methodology. We also conducted an exploratory analysis to identify the factors behind cluster formation. The insights gained can be extrapolated to analogous experiments where social or economic data are scarce, and most of the available attributes are academic in nature.

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Turbofan engine health status prediction with heterogeneous ensemble deep neural networks

Información de publicación: Szrama, Slawomir.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Aircraft turbofan engines undergo gradual performance degradation that, if left undetected, can compromise flight safety and increase maintenance costs. In this study, we propose a heterogeneous ensemble deep neural network (HEDNN) for multi-class health status prediction of real-life turbofan engines. Our methodology begins with the collection of 43,492 real operational records (2012–2024) from a dual-spool, variable geometry turbofan, each run sampled at a fixed throttle setting. We preprocess raw sensor outputs using a robust z-score (median absolute deviation) filter to remove outliers, apply a 35-sample moving median imputation to address 1,812 missing temperature entries, and eliminate any non-numeric artifacts. The cleaned dataset consisting of 97 predictive features after FADEC upgrade harmonization is partitioned into training (80%, n=34 794) and testing (20%, n=8 698) sets, and labeled into three health classes (no/low degradation, mid-life degradation, and advanced degradation) built upon performance margin (PMAR) and specific fuel consumption (SMAR) indices.

We engineer three base learners: bidirectional LSTM (128 hidden units), one-dimensional CNN (11 convolution normalization ELU blocks), and BiGRU (four GRU layers, custom flip/concatenate stages), each trained for up to 60 epochs, batch size 16, using the Adam optimizer with piecewise learning rate decay (initial η=0.01). Individual networks produce

class probability vectors that are fused in a shallow fully connected ensemble, finalized by softmax classification. Performance is assessed via RMSE, MAPE, overall accuracy, precision, recall, F1 score, and multi-class AUC. The HEDNN achieves RMSE=0.3421 (33–41% lower than any single model), MAPE=4.78% (60–65% reduction), and accuracy=95.32%, with F1=0.928 and mean AUC=0.777. In contrast, the best individual learner (BiLSTM) reaches RMSE=0.5149, MAPE=12.17%, accuracy=82.44%, and AUC=0.717. These results demonstrate that our heterogeneous ensemble markedly outperforms standalone architectures, offering a robust, data-driven prognostic tool for predictive maintenance of turbofan fleets.

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Gower-SOM: a self-organizing map for mixed data with gower distance and heuristic adaptation for data analytics

Información de publicación: Sáez, Patricio; Salas, Patricio.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Self-organizing maps (SOMs) remain a benchmark technique for low-dimensional representation and cluster discovery; however, the original algorithm assumes Euclidean geometry and is therefore ill-suited to datasets that combine numerical, binary, and categorical features. This study introduces Gower-SOM, a topology-preserving neural network that replaces the Euclidean metric with Gower distance and incorporates probabilistic update rules for non-numeric features. The algorithm processes mixed-type data without one-hot-encoding or hierarchical transformations, thus avoiding dimensional inflation and semantic distortion. We evaluate Gower-SOM on three synthetic benchmarks designed to stress feature heterogeneity and on two publicly available datasets: Adult (approximately 48,000 instances, 14 features) and Obesity (approximately 2100 instances, 17 features). In comparison to the classical SOM, the proposed model reduces the normalized quantization error by 75–99% in synthetic scenarios and by 87% and 68% on the Adult and Obesity datasets, respectively. Cluster validity improves concurrently: the average silhouette index rises from 0.31 to 0.50 (adult) and from 0.43 to 0.48 (obesity), indicating minimal distortion of neighborhood relations. U-Matrix visualizations reveal sharper boundaries and more coherent attribute gradients, underscoring superior interpretability. These results demonstrate that Gower-SOM improves the mapping of mixed-type data without increasing algorithmic complexity, positioning it as a rigorous and scalable alternative for exploratory analysis, feature engineering, and decision support in domains where mixed-type data are the norm.

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Biomedical text summarization with large language models: methodologies, challenges, and future directions

Información de publicación: Mushtaq, Shafiya; Veningston, K.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

The exponential growth of biomedical literature, clinical records, and health-related publications necessitates robust and scalable methods for automatic text summarization. Recent advancements in large language models (LLMs), such as BioBERT, PubMedBERT, and Med-PaLM, have significantly enhanced the capabilities of natural language processing systems to generate concise, coherent, and contextually relevant summaries tailored to the biomedical domain. This survey provides a comprehensive review of state-of-the-art methodologies for biomedical text summarization using LLMs, encompassing both extractive and abstractive approaches. We systematically examine model architectures, training paradigms, domain-specific adaptations, and evaluation techniques across various biomedical subdomains. The review adheres to the PRISMA methodology, ensuring a rigorous and reproducible literature selection process from 2016 to 2024, drawing from reputed databases including PubMed, Scopus, IEEE Xplore, and ACM Digital Library. In addition to synthesizing recent advances, the paper critically outlines prevailing challenges that impede the deployment of LLMs in biomedical summarization tasks. These include the complexity of domain-specific vocabulary and abbreviations, risks of hallucination and factual inconsistency, limited availability of high-quality annotated datasets, inadequacy of standard evaluation metrics in capturing clinical relevance, and ethical concerns surrounding data privacy and bias. Addressing these challenges is imperative to advancing the development of accurate, reliable, and ethically sound summarization systems for biomedical applications.

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Data-driven analysis and prediction of traffic accident dynamics using spatiotemporal modeling and optimized machine learning techniques

Información de publicación: Turab, Doha; Muzaffar, Sharjeel; Nadeem, Lubna; Li, Jianqiang; Wang, Yu; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Road traffic accidents represent a significant safety concern worldwide, ranking among the leading causes of fatalities and a major risk to human life and property. Addressing this challenge requires a comprehensive understanding of the factors contributing to accidents and their consequences. This study uses advanced data mining and analysis techniques to investigate data from traffic accidents, focusing on temporal, spatial, environmental, and road-related factors. The research aims to uncover hidden patterns within the data and provide a scientific basis for proactive accident prevention and efficient post-accident response. The study begins with a comprehensive preprocessing workflow, including data reduction, transformation, and cleaning, ensuring the validity and reliability of the data. Temporal, spatial, and environmental characteristics, such as time, location, weather, temperature, and lighting, are analyzed and visualized to understand traffic accident distribution patterns comprehensively. An association analysis is then performed using the *K*-means clustering algorithm to discretize continuous accident variables, followed by applying the Apriori algorithm to uncover hidden relationships among accident causes. An accident severity-weighted quantization method enhances the

algorithm, effectively identifying significant association rules related to severe accidents. Based on these findings, targeted accident prevention strategies are proposed. Finally, the study evaluates multiple deep learning and machine learning algorithms for accident duration prediction, identifying optimized support vector regression via Bayesian and particle swarm optimization (BPSO-SVR) as the best-performing model. The optimized BPSO-SVR model combines Bayesian optimization (BO) and particle swarm optimization (PSO) to fine-tune model hyperparameters, significantly improving prediction accuracy. An interpretability analysis identifies key factors influencing accident duration, including severity, crossing characteristics, sunrise/sunset conditions, and weather. These findings contribute to advancing road traffic safety research, offering valuable insights for early-warning systems and efficient post-accident interventions.

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Logistic Dropout: a soft dropout mechanism via the Logistic Map

Información de publicación: Brezinski, Kenneth; Ferens, Ken; Kinsner, Witold.

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RESUMEN (ENGLISH)

We propose *Logistic Map Dropout*, a novel regularization mechanism that leverages the chaotic dynamics of the logistic map to perform continuous, stateful dropout in deep neural networks. Unlike standard dropout which stochastically zeros out activations, our method deterministically scales activations by a value in [0, 1] derived from the logistic map, enabling smooth attenuation of activations and uninterrupted gradient flow. To bridge the gap between continuous and binary regularization, we introduce two extensions: *Sparse Scaling*, which thresholds the logistic map to induce sparsity while preserving strong activations, and *Modulated Sparse Scaling*, which allows selective zeroing to approximate classical dropout behavior. Extensive empirical evaluations across image classification, language modeling, and graph prediction tasks demonstrate that our method achieves competitive or superior generalization compared to standard and structured dropout baselines. In particular, modulated variants consistently outperform stochastic regularizers on LSTM and GNN architectures, highlighting the benefit of chaos-driven deterministic masking. Logistic Map Dropout offers a principled alternative to stochastic regularization by blending chaotic dynamics with structured noise, yielding a tunable spectrum between soft and hard dropout regimes. This work opens a new direction for applying dynamical systems theory in neural network regularization.

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Explainable deep learning for software defect prediction: a residual-shuffle network approach with SHAP and LIME

Información de publicación: Talatapeh, Samira Keramat; Dishabi, Mohammad Reza Ebrahimi; Pezeshki, Hamed; Madadi, Halimeh.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Software defect prediction (SDP) is essential for maintaining software quality, especially in critical systems like telecommunications. However, the lack of transparency in deep learning models hampers their practical application. This paper introduces an explainable deep learning framework for software defect prediction (SDP), combining a residual-shuffle network (RSN) with two model-agnostic interpretability tools—Shapley additive explanations (SHAP) and local interpretable model-agnostic explanations (LIME)—to enhance the explainability of predictions using explainable artificial intelligence (XAI). The RSN architecture is designed to capture complex software patterns through residual learning and channel shuffling, while SHAP and LIME provide interpretable insights into individual predictions and global feature importance. We evaluate the framework on public benchmark datasets (PROMISE/NASA) and industrial datasets from the telecommunications sector. Results show that the RSN+XAI model outperforms traditional models in terms of accuracy, precision, recall, and F1-score. Additionally, user feedback from industrial practitioners reveals that the interpretability of the model aids in resource allocation, debugging, and meeting regulatory compliance. This work demonstrates the importance of model transparency in SDP and paves the way for future research on real-time feedback mechanisms and the integration of natural language features to further improve model explainability in software engineering contexts.

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A systematic review of plant leaf disease detection and classification using machine learning and deep learning techniques

Información de publicación: Kumari, Vinesh; Bansal, Varun.

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RESUMEN (ENGLISH)

Plant health in agriculture is very important for crop productivity and food security. Conventional methods of plant disease detection are manual, requiring loads of time and labor, and are often subjected to human error. Considering that the visual manifestations of leaf disease are one of the basic symptoms of diseases, which can be promoted or detected first on leaves, the focus of major research has shifted toward leaf disease detection using computer vision. In computer vision, deep learning models achieved a superior performance showing an ideal solution to plant diseases diagnosis from an image-based leaf analysis. A detailed study was conducted from 2015 to 2025, which highlights the features of advanced techniques in plant leaf disease detection. Under this study, a systematic review and evaluation of advanced machine learning and deep learning methods which are used for recognition and classification of plant leaf diseases is conducted. Amid all the models, convolution neural network is the most accurate model in detecting plant leaf diseases. The major focus of this review is to guide the researchers to identify the most efficient approaches for plant leaf disease detection. It also lists the advantages and drawbacks of current models that are based on these models. Nine major

research inquiries were prepared to govern the review's scope; these questions covered aspects including datasets, image preprocessing, image segmentation, feature extraction, and classification techniques. It concludes by providing relevant answers to the research inquiries posed, in addition to future recommendations for plant leaf disease detection and classification.

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Interactive story maps for historical musical instruments: a 3D and semantic web tool for cultural heritage preservation

Información de publicación: Bartalesi, Valentina; Lenzi, Emanuele; De Martino, Claudio; Coro, Gianpaolo.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Ancient stringed instruments from the 17th and 18th centuries are essential for global cultural heritage but pose preservation and study challenges due to their fragility and rarity. Digital reproductions and interactive story maps offer new ways to analyse these instruments, providing historical, mechanical, and acoustic insights while aiding modern luthiers in replication. This paper presents a novel methodology and software that integrates 3D models of historical instruments into interactive story maps using Semantic Web technologies. The system enables users to explore 3D models with embedded annotations from Sketchfab and supports the creation of formal narratives enriched with geospatial and multimedia content. Each narrative event is linked to a backend knowledge base (KB) structured with established ontologies. The tool is open-source and web-based, and allows interoperability between its story maps and external KBs such as Wikidata and Europeana. It supports multiple levels of descriptive details, from precise 3D model annotations to broad geographic and temporal references. An evaluation case study featured 25 participants exploring a story map about a 1737 Antonio Stradivari violin, connected to a KB of 1716 triples. The results highlight the tool's usability and effectiveness in representing the violin's spatiotemporal and cultural context, showcasing the value of combining 3D models and geospatial data to enhance cultural heritage preservation.

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Forecasting refugee migration with high-dimensional covariate space

Información de publicación: Qi, Haodong; Sîrbu, Alina; Momeni, Rahman; Hisam, Enes; Arcila-Calderón, Carlos; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Forecasting refugee migration is challenging, exacerbated by the high dimensional and dynamic nature of its drivers, such as climatic, economic, and political stressors. This article introduces a novel forecasting framework based on the Dynamic Elastic Net (DynENet) algorithm, which incorporates a time-varying regularization and a new model selection criterion: the penalized deviance ratio (PDR). Unlike conventional metrics such as the deviance ratio (DR), which emphasize in-sample fit, PDR explicitly penalizes model complexity, enhancing generalization in high-dimensional covariate setting. We apply this framework to forecast asylum-seeker rates (ASR) from Somalia to EU member states, leveraging a comprehensive set of district-level predictors. Extensive validation demonstrates that PDR-tuned models consistently outperform DR-based benchmarks in out-of-sample accuracy, reducing average point prediction errors by 40% and improving interval forecasts by 79%. Furthermore, we demonstrate how the DynENet framework supports explanatory insights at multiple levels—origin district, destination, and temporal—revealing both persistent and transient nature of migration drivers. The proposed methodology not only advances forecasting accuracy under high-dimensional covariate conditions but also enhances the interpretability of complex and evolving migration systems.

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Hierarchical growth self-generating prototype: an information entropy-based approach for dataset reduction

Información de publicación: Manastarla, Alberto; Silva, Leandro A.

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RESUMEN (ENGLISH)

In instance-based learning algorithms, storing numerous training examples creates high memory demands, sensitivity to noise, and slower execution. Instance selection addresses these issues by retaining a compact subset that removes redundancy and noise while preserving essential structure. Because the value of a training example depends on its neighborhood, reduced sets should include representatives from safe and ambiguous subregions. This work introduces the Hierarchical Growth Self-generating Prototype (HSGP), an entropy-guided instance-selection algorithm framed through decision-tree principles. HSGP recursively partitions the data by impurity, grows low-uncertainty regions, and synthesizes provisional prototypes. Each prototype serves only as an anchor and is replaced by its nearest real training instances, preserving local density and label fidelity while avoiding artifacts from synthetic points. To prevent over-refinement, HSGP applies a stopping rule based on the dispersion of subset entropies; partitioning halts when uncertainty becomes uniformly low across the active subsets, curbing unnecessary splits and maintaining a favorable balance between reduction and accuracy. The algorithm selects examples from distinct and overlapping class regions, capturing dataset complexity while retaining interpretability. Experiments on twenty-one real-world classification problems show competitive or superior predictive accuracy relative to 1-Nearest Neighbor trained on the full dataset and favorable performance against established instance-selection methods, while delivering substantial reductions in set size. Overall, entropy-guided growth with anchor-and-replace selection offers an effective strategy for building compact training sets that preserve decision-relevant structure.

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An explainable AI-driven framework integrating SHAP and QFD for transitioning from CRM to CJM in customer-centric optimization

Información de publicación: Wang, Tzu-Chien.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Customer journey management (CJM) has become increasingly important for firms transitioning from traditional customer relationship management (CRM), requiring analytical methods that both explain behavioral drivers and translate insights into operationally executable plans. This study proposes an explainable and operations-ready framework that integrates the pre-purchase and purchase stages, with an optional post-purchase extension via product-review analytics. The framework combines clickstream-based clustering, latent Dirichlet allocation (LDA) for mapping customer-need topics to product functions and specifications, and tree-based gradient-boosting models with Shapley additive explanations (SHAP) to identify the key determinants of click engagement, click-to-purchase behavior, and transaction value. These explainable outputs are transformed into a quality function deployment (QFD) matrix, which is subsequently incorporated into a mixed-integer linear programming (MILP) allocator constrained by budget, delivery capacity, minimum-coverage requirements, and frequency-capping rules. The empirical analysis uses enterprise-scale behavioral and transactional data from a leading Taiwanese consumer electronics firm, comprising 412,580 users and 156 source/medium pairs (March–November 2020), supplemented by 3,492,632 product-review records (January 2021–August 2022). The results reveal substantial cross-segment heterogeneity in customer value and reachable audiences, supporting the need for differentiated segment–channel strategies. In the review-analytics extension, an LDA–LightGBM configuration achieves predictive performance comparable to LSTM and CNN baselines while offering significantly faster training, enabling more frequent model updates. The MILP produces feasible and interpretable segment–channel allocations, validated through return on ad spend (ROAS), cost per acquisition (CPA), and segment-level conversion-lift analysis. Overall, the integrated SHAP–QFD–MILP pipeline demonstrates how explainable modeling outputs can be operationalized into prescriptive and auditable decisions, providing a scalable blueprint for customer-centric resource planning across products, markets, and channels.

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Overfitting, regularisation and condition estimation in regression

Información de publicación: Winkler, Joab R.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Overfitting is a problem in regression and deep neural networks, and it is often stated that Tikhonov regularisation minimises its adverse effects, but the relationship between regularisation and overfitting has not been established. The theory of regularisation is well developed, but overfitting has a qualitative description and it is not defined mathematically. This paper addresses the relationship between overfitting, regularisation and condition estimation by considering underdetermined and overdetermined least squares (LS) problems that arise in regression. This study is important because regularisation is not benign since its use when a condition on the decay of the singular values of the coefficient matrix in the LS minimisation is not satisfied leads to a large error in the solution of the regularised LS problem. Examples in which the regression curve overfits the data are shown, but regularisation must not be applied because the LS problem is well conditioned. Also, an ill conditioned LS problem whose solution does not display overfitting is shown, but its ill conditioned nature implies regularisation should be applied in order to obtain a numerically stable solution. It is concluded that regularisation does not solve the problem of overfitting in regression.

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Deep learning for new fashion product demand prediction: integrating visual similarity and demand correction in cold-start scenarios

Información de publicación: Adel, Mona; Mohammed, Amgad M; Elsaid, Asmaa H; Abdelatey, Amira.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Forecasting demand for new fashion items poses a persistent challenge in retail analytics, primarily due to the unavailability of historical sales data. This study presents a hybrid deep learning architecture tailored to address the cold-start problem by combining visual similarity analysis with correction techniques for censored sales data. The proposed approach utilizes multimodal inputs, including high-resolution product images and structured metadata (e.g., category, fabric, and price), in addition to sales and inventory records that are affected by stockout-related censoring. The architecture is composed of two integrated pipelines. The first applies the Expectation–Maximization (EM) algorithm to adjust sales records, estimating true latent demand by accounting for censored entries. The second pipeline extracts deep visual features through a modified ResNet-50 architecture, followed by compression via an autoencoder to generate compact, semantically meaningful representations. These representations are clustered to identify visually similar products, enabling demand estimation through similarity-weighted averaging and localized regression within each cluster. Prediction uncertainty is captured using a bootstrap-based resampling technique. This work offers several notable contributions: the fusion of EM-based demand correction with deep visual learning, the introduction of an adaptive similarity-based forecasting strategy, and the integration of uncertainty estimation for cold-start scenarios. Empirical evaluation on a real-world multimodal fashion dataset demonstrates that the proposed model significantly outperforms several baseline approaches—including those relying solely on metadata, visual features, or global regressors—with results achieving RMSE of 10.95, MAE of 8.54, MAPE of 16.5%, Precision@10 of 73.9%, and Recall@10 of 70.7%. These outcomes underscore the effectiveness and practical value of the architecture in predicting demand for newly launched fashion products.

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Exploratory analysis of logical and intuitive reasoning in large language models

Información de publicación: Peruchini, Melise; Monteiro Teixeira, Júlio.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This study consists of qualitative empirical research conducted through exploratory tests with two large language models (LLMs): ChatGPT and Gemini. The methodology involved exploratory tests based on prompts designed with a probability question, using the well-documented “Linda Problem,” from cognitive psychology, and a newly developed “Mary Problem.” The analysis focuses on the outputs generated by each chatbot to assess whether their reasoning aligns with probability theory or is influenced by stereotypical descriptions in the prompts. The findings provide insights into how each chatbot handles logic and textual constructions, suggesting that, while both performed adequately on a well-known probabilistic problem, they exhibit significantly lower performance on novel tests requiring direct application of logic involving probabilities.

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SentiVol-GA: a volatility-scaled genetic fusion of predictive models and financial sentiment for adaptive stock forecasting

Información de publicación: Singh, Monika; Tunga, Harinandan; Kar, Samarjit.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

SentiVol-GA is an evolvable hybrid framework for stock price forecasting. This model integrates statistical models, deep learning architectures, financial sentiment analysis, and volatility-aware optimization through a Genetic Algorithm (GA). This framework combines five predictive models—Linear Regression, LSTM, GRU, Bi-LSTM, and ARIMA—with sentiment signals extracted from FinBERT, VADER, and the Loughran–McDonald dictionary. Its core innovation lies in a volatility-scaling mechanism that adjusts sentiment impact during turbulent periods, while the GA periodically optimizes model and sentiment weights to minimize forecasting error thus sustaining the model’s predictive stability. This empirical evaluation across eight Indian IT-sector stocks—covering large, mid- and small-cap segments, demonstrates up to 12% improvement in R^2, 30–60% reduction in RMSE, and 20–35% gain in tolerance-based accuracy over classical and deep-learning baselines.

Nonparametric Friedman and Wilcoxon tests ($p<0.05$) confirm that these performance gains are statistically significant. SentiVol-GA offers a volatility-aware, sentiment-driven, and dynamically optimized approach that is scalable, interpretable, and robust for intelligent financial decision-making.

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Inf-DLFS: integrating deep learning with optimal feature selection to identifying influential nodes in complex networks

Información de publicación: Singh, Naveen Kumar; Sharma, Saurabh Kumar.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Identifying influential nodes in complex networks is a fundamental problem with significant applications in areas such as information diffusion, epidemic control, and social network analysis. This research proposes a new deep learning-based framework, Inf-DLFS, which integrates deep feature engineering, optimization, and dynamic propagation modeling to accurately identify influential nodes. The method comprises three main components. First, a multistage artificial neural network (MS-ANN) is employed to extract informative node-level features by leveraging structural properties like degree, closeness, betweenness, and eigenvector centrality. This produces robust feature representations embedded with neighborhood context. Second, an improved selfish herd optimization (ISHO) algorithm is used for optimal feature selection. ISHO iteratively evaluates feature subsets using K-means clustering and silhouette scoring, thereby reducing dimensionality and selecting the most relevant features for influence prediction. Finally, the influence scores are computed using the impulsive pantograph neural network (IPNN), which models influence propagation through time with delayed feedback and periodic impulses. The proposed IPNN dynamically updates node states based on historical interactions and graph topology over multiple time steps. Extensive experiments on real-world datasets demonstrate that Inf-DLFS significantly outperforms recent deep learning methods in terms of accuracy, stability, and scalability. The integration of graph-based structural features, bio-inspired optimization, and dynamic modeling makes Inf-DLFS a powerful solution for influential node identification in complex networks.

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A parametric distributional reinforcement learning framework for conditional systemic risk estimation

Información de publicación: Leballo, Keorapetse; Mba, Jules Clement.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

We extend the distributional reinforcement learning framework by incorporating parametric tail modeling via the generalized extreme value (GEV) distribution into the value estimation process, augmented with superior learning enhancements. This allows for principled modeling of systemic risk measures such as the System Conditional Value-at-Risk (S-CoVaR) and enables risk-sensitive policy learning in environments associated with heavy-tailed reward distributions, such as asset class returns. Additionally, we explore nonparametric empirical distribution modeling to provide a flexible alternative and evaluate the agents' estimated results across bull and bear market conditions. Our research findings highlight the vital role that distributional assumptions play in frameworks geared towards risk-sensitive decision-making under economic and financial market uncertainty.

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Arabic word sense disambiguation: a survey in the era of transformer-based models

Información de publicación: Saidi, Rakia; Jarray, Fethi.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

This survey reviews advancements in Arabic Word Sense Disambiguation (WSD), with a focus on contextualized methods such as deep learning and transformer models. Despite the linguistic complexities of Arabic and the scarcity of annotated datasets, WSD remains a critical challenge in Natural Language Processing (NLP). This paper systematically examines traditional knowledge-based methods, machine learning approaches, deep learning models, and hybrid techniques, with a particular emphasis on transformer models such as BERT, which have recently achieved state-of-the-art performance in Arabic WSD. The survey outlines the criteria for model selection, including performance on benchmark datasets and relevance to real-world applications. It also provides a comparative evaluation of the methodologies, highlighting their strengths and weaknesses through structured comparisons of Accuracy, Precision, Recall, and F1-score. Unlike previous surveys, this work offers a comprehensive analysis that integrates multiple evaluation metrics across diverse studies. The paper concludes by identifying key challenges, such as the need for larger annotated datasets and improved contextual understanding, while proposing future research directions, including domain adaptation and cross-dialectal WSD.

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CDGRS-LOD model: leveraging linked open data in collaborative cross-domain group recommender system

Información de publicación: Xuan, Yui Chee; Nawi, Rosmamalmi Mat; Noah, Shahrul Azman Mohd; Mahdin, Hairulnizam.

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RESUMEN (ENGLISH)

This study introduces a novel Cross-Domain Group Recommender System leveraging Linked Open Data (LOD) (CDGRS-LOD), designed to improve the accuracy and relevance of group recommendations. Group recommender systems (GRS) are increasingly used to deliver collective recommendations, but their effectiveness is often constrained by sparse user data and limited semantic context during group formation. The proposed CDGRS-LOD model enriches movie metadata with structured external knowledge, incorporating cross-domain attributes such as the Music Composer feature extracted from DBpedia via SPARQL queries. This enhancement improves the model's understanding of relationships between users and items, thereby enabling more accurate and contextually relevant group recommendations. Singular Value Decomposition (SVD) predicts missing ratings from the enriched metadata, effectively mitigating sparsity. Two sparsity reduction levels, 1% and 5%, were evaluated, with the 5% reduction yielding the best performance and subsequently used for group formation with a k-Nearest Neighbours (kNN) model. The model is evaluated against existing methods, including GRS-LOD, Archi2_bkm, Archi2_km, and Predict & Cluster. Results show that CDGRS-LOD achieves a Root Mean Square Error (RMSE) of 0.5917, outperforming GRS-LOD (0.7816) and traditional approaches (0.9435–0.9781), representing up to 24% improvement over the best baseline. These findings confirm that integrating cross-domain knowledge with LOD enhances metadata richness, reduces sparsity, lowers prediction errors, and improves group recommendation quality.

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Machine learning for optimized crop selection: a data-driven approach considering climate and soil type data

Información de publicación: Gomes da Costa Cavalcanti, Joao Henrique; de Brito Souto, Moisés Cirilo; de Moura Oliveira Filho, Eduardo; Rodrigues, Erich Matos; Barros, Yan Evangelista; et al.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

Agriculture is vital for food security and economic development and optimizing crop selection based on data is key for sustainable farming. This study uses machine learning (ML) models to recommend suitable crops based on climate and soil data in Brazil, addressing a gap in the literature that often focuses on Kaggle data, localized data and individual soil components. Using data from the Brazilian Ministry of Agriculture and the National Institute of Meteorology (INMET), the study aggregates crop productivity data from 32 soil types and 95 crop types farmed in 3961 cities in Brazil with daily climate data. Various ML techniques were explored, including Gradient Boosting, Logistic Regression with Bagging, support vector machine (SVM) with Bagging, and Random Forest, all the ML methods tested different preprocessing methods. Synthetic Minority Over-sampling Technique (SMOTE) addressed class imbalance, and multithreading improved computational efficiency. Results showed that the Random Forest performed best with climate data alone (F1-score of 0.20), while adding soil types increased the score to 0.42. For city-specific data, Gradient Boosting achieved an F1-score of 0.70. Tailored models for specific cities outperformed generalized ones, highlighting the challenges of creating general

models. Localized models demonstrated to be easier to train but have less generalizing power.
Graphic abstract

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Generative conditional missing imputation networks

Información de publicación: Sun, George; Zhou, Yi-Hui.

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RESUMEN (ENGLISH)

In this study, we introduce a sophisticated generative conditional strategy designed to impute missing values within datasets, an area of considerable importance in statistical analysis. Specifically, we initially elucidate the theoretical underpinnings of the Generative Conditional Missing Imputation Networks (GCMI), demonstrating its robust properties in the context of the Missing Completely at Random (MCAR) and the Missing at Random (MAR) mechanisms. Subsequently, we enhance the robustness and accuracy of GCMI by integrating a multiple imputation framework using a chained equations approach. This innovation serves to bolster model stability and improve imputation performance significantly. Finally, through a series of meticulous simulations and empirical assessments utilizing benchmark datasets, we establish the superior efficacy of our proposed methods when juxtaposed with other leading imputation techniques currently available. This comprehensive evaluation not only underscores the practicality of GCMI but also affirms its potential as a leading-edge tool in the field of statistical data analysis.

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Strategic management of human skills in Big Data initiatives: from SLR to skills taxonomy and human resource management framework

Información de publicación: Totaro, Nicolò Gianmauro; Specchia, Giorgia; Corallo, Angelo; Gervasi, Massimiliano.

Enlace de documentos de ProQuest

RESUMEN (ENGLISH)

In the context of accelerating digital transformation, the alignment between human capital and technological innovation has become a strategic priority for organisations operating in data-intensive environments. However, the literature still exhibits significant fragmentation in the classification of the human skills required in this domain, and operational tools for their strategic management remain limited. Guided by the theoretical frameworks of the resource-based view and the

dynamic capability view, this study addresses this dual gap through a systematic literature review. First, it presents a multi-level taxonomy of human skills relevant to Big Data initiatives. The taxonomy organises hard, managerial, and soft skills into a structured classification that serves both theoretical consolidation and practical applications. Second, the study introduces the AFE (awareness, fulfilment, efficiency) framework, a conceptual model designed to support the dynamic assessment and strategic optimisation of human capital. In addition to traditional maturity models, AFE incorporates subjective dimensions, such as self-awareness and job satisfaction, allowing organisations to identify discrepancies between perceived competencies, actual skill usage, and role expectations. Together, the taxonomy and the AFE framework offer a comprehensive foundation for improving skill assessment, guiding workforce planning, and supporting reskilling and job redesign initiatives. This work contributes to bridging the gap between descriptive literature and actionable tools, fostering more adaptive, evidence-based, and human-centred approaches to human resource management in Big Data contexts.

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Continuous software monitoring backed by process mining: a systematic literature review

Información de publicación: Stepanov, Evgenii; Mitsyuk, Alexey.

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RESUMEN (ENGLISH)

Software systems are monitored constantly, as it is the only way to ensure their well-functioning. There are several approaches for software monitoring: starting with debugging and profiling of simple programs, and ending with large distributed systems which are monitored by a complex logging infrastructure. As a result of such a monitoring, aggregated numbers (i.e., the average amount of memory or CPU used) and charts (i.e., the number of web requests per minute) are produced. In this paper, we will focus on another way of software monitoring, which uses process mining under the hood. Firstly, we introduce the desired architecture of the process mining monitoring system, then identify the fields of process mining that should be researched, and finally conduct a systematic literature review in order to understand what approaches can be used in continuous process mining monitoring scenarios.

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Methods and evaluation in unsupervised keyphrase prediction: a survey

Información de publicación: Han, Yuchen; Wu, Huiqian; Sun, Yuqing.

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RESUMEN (ENGLISH)

Keyphrases are concise semantic units that capture the meanings of a document, providing interpretable abstractions that reduce textual complexity and support downstream applications such as information retrieval, text summarization, and document classification. While supervised keyphrase prediction methods rely on costly annotated data, unsupervised methods have emerged as effective alternatives. This survey provides a comprehensive overview of unsupervised keyphrase prediction methods and systematically reviews the evaluation metrics. We analyze the linguistic properties of keyphrases to inform the design of evaluation metrics and guide the improvement of prediction methods. In addition, we categorize existing unsupervised approaches and discuss their respective strengths and limitations. Furthermore, we review existing evaluation metrics and analyze their shortcomings. To address these gaps, we introduce CAME, a novel reference-free metric, and validate its effectiveness through experiments. Finally, we highlight challenges and promising directions, particularly in the context of LLMs, to guide future research in unsupervised keyphrase prediction.

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Interpretable and robust multimodal fake news detection with evidence fusion and reliable reasoning

Información de publicación: Thakur, Rahul; Kavindu Jayasinghe, J. P. Teshan; Mehta, Preeti.

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RESUMEN (ENGLISH)

Multi-modal fake news detection involves the complex task of jointly analyzing textual and visual information, whereas uni-modal systems often under-perform due to limited contextual understanding. This paper presents a reliable and interpretable multi-modal framework that integrates semantic and stylistic cues using BERT for textual encoding, and CLIP and SWIN Transformer models for visual representation. Intra-modal features are fused via a Dempster–Shafer theory-based gating mechanism, enabling uncertainty-aware reasoning within each modality. Cross-modal features are aligned and refined using transformer-based correlation modeling to capture nuanced interactions between modalities. The framework is further strengthened by a composite loss function, formed by combining evidential deep learning with supervised and unsupervised contrastive learning strategies. This results in improving the classification accuracy and enhancing the model's robustness by quantifying uncertainty and allowing the rejection of low-confidence predictions. The inclusion of the uncertainty regularization ensures well-calibrated outputs, which are crucial for real-world deployment where interpret-ability and trust are essential. Extensive experiments on the Fakeddit dataset demonstrate that the proposed approach consistently outperforms strong baselines in terms of accuracy, uncertainty estimation, and transparency of decisions. By tightly coupling multi-modal evidence integration with principled uncertainty modeling, the framework advances the state of the art in trustworthy misinformation detection.

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Taurus vs stairway to heaven: a novel MIDI-note empty-space dataset algorithm to detect music plagiarism

Información de publicación: Howard, Jeffrey N.

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RESUMEN (ENGLISH)

The current study highlights application of a novel statistical algorithm as proof-of-concept in detecting musical passage similarity via the construction of a unique difference-score dataset, comprised of MIDI-note (Musical instrument digital interface) value differences, and the 'empty spaces' (pixel distances) between printed musical score notation elements. Using the well-known case of Skidmore vs Led Zeppelin (Leo, J Am Musicol Soc 77(2):303–333, 2024), brief musical scores of the songs 'Taurus' and 'Stairway to Heaven' were compared via bootstrapping (resampling) analysis of a uniquely engineered difference-score dataset. Results indicate no-difference between the musical scores of interest, thus supporting a 'plagiaristic' outcome when analysis is based solely on quantitative data values derived from MIDI-note data, and temporal physical space characteristics between notes on a musical score. In addition, six other well-known plagiarism cases, including former Beatle George Harrison's (1970) song 'My Sweet Lord' vs 'He's So Fine' by The Chiffon's (1963), a plagiarism case from 1981 (Perot, Berkeley J Entertain Sports Law, 14, 2025), are also subjected to the novel MNV-ESV algorithm.

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Symmetry group equivariant convolutions for representation learning: a survey

Información de publicación: Basheer, Ramzan; Mishra, Deepak.

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RESUMEN (ENGLISH)

Euclidean deep learning is often inadequate for addressing real-world signals where the representation space is irregular and curved with complex topologies. Interpreting the geometric properties of such feature spaces has become paramount in obtaining robust and compact feature representations that remain unaffected by non-trivial geometric transformations, which vanilla CNNs cannot effectively handle. Recognizing rotation, translation, permutation, or scale symmetries can lead to equivariance properties in the learned representations. This has led to notable advancements in computer vision and machine learning tasks under the framework of geometric deep learning, as compared to their invariant counterparts. In this report, we emphasize the importance of symmetry group equivariant deep learning models and their realization of convolution-like operations on graphs, 3D shapes, and non-Euclidean spaces by leveraging group theory and symmetry. We categorize them as regular, steerable, and PDE-based convolutions and thoroughly examine the inherent symmetries of their input spaces and ensuing representations. We also outline the mathematical link between group convolutions or message aggregation operations and the concept of equivariance to serve as a valuable reference and stimulate further research in this emerging discipline.

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A layer-wise survey on internal modifications in BERT and its variants: techniques, applications, and performance trade-offs

Información de publicación: Saxena, Aradhana; Santhanavijayan, A.

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RESUMEN (ENGLISH)

As the volume of unstructured text grows, transformer models like BERT have become cornerstones of natural language processing (NLP). Yet, researchers increasingly recognize that a one-size-fits-all BERT may not optimally serve every task or domain. In response, a wide array of layer-wise customizations has emerged, where components of BERT (from embeddings to output heads) are redesigned for specific needs. In this survey, we present a comprehensive layer-by-layer analysis of over 90 BERT variants (2021–2025), highlighting architectural innovations across the embedding, attention, feed-forward, normalization, and output layers. To organize this rich landscape, we introduce a structured taxonomy that links each modification to its target task and dataset. We discuss techniques such as sparse attention for speedup, multimodal embeddings for richer context, and privacy-aware classifiers for secure inference. Reported results are striking: For example, some embedding-level changes achieve 99.94% accuracy on sentence similarity with 3× model compression, while hybrid attention designs have pushed medical named entity recognition (NER) accuracy as high as 94%. We place these advances in context with comparative tables and timeline figures that illustrate the trade-offs between accuracy, efficiency, and applicability. Crucially, we connect the dots to real-world impact: For instance, domain-tuned BERT models have powered clinical decision support by extracting patient information from electronic health records (boosting disease prediction accuracy by up to 6%), and financial sentiment analysis by classifying market news with fine-tuned attention heads. By synthesizing these trends, our survey provides a modular, application-aware perspective on BERT's evolution, serving as a practical reference for researchers and practitioners building optimized transformer models.

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